

French Study Abroad Project Analysis

```
# Load libraries
library(MASS)
library(car)
```

Loading required package: carData

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.2.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
x dplyr::recode() masks car::recode()
x dplyr::select() masks MASS::select()
x purrr::some()   masks car::some()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Load Data

```
# Load the preprocessed data
french_data <- read_csv("data/Processed_French_Data.csv",
                        show_col_types = FALSE)

french_data
```

```
# A tibble: 58 x 18
  Year_2024 Year_2025 Class_Level Host_Family Phone_Language Phone_Keyboard
    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1         0         0         2         0         1         1
2         0         0         2         0         1         1
3         0         0         0         0         1         1
4         0         0         2         0         1         1
5         0         0         3         1         1         1
6         0         0         0         0         0         1
7         0         0         3         0         0         1
8         0         0         4         1         0         1
9         0         0         0         0         0         1
10        0         0         1         1         1         1
# i 48 more rows
# i 12 more variables: Screen_Time <dbl>, Interaction_Score <dbl>,
#   Speaking_Score <dbl>, Speak_Hours_Score <dbl>, Use_French_Score <dbl>,
#   French_Hours_Score <dbl>, Read_French_Score <dbl>,
#   Write_French_Score <dbl>, Listen_French_Score <dbl>,
#   Speak_French_Score <dbl>, Competence_Score <dbl>, Intention_Score <dbl>
```

EDA

Summary Statistics

The data has 58 observations and 14 variables that we will be analyzing. Twelve of these variables are predictors and two, Competence_Score and Intention_Score, are response variables.

```
# Check for missing data
colSums(is.na(french_data))
```

Year_2024	Year_2025	Class_Level	Host_Family
0	0	0	0
Phone_Language	Phone_Keyboard	Screen_Time	Interaction_Score

	0	0	31	0
Speaking_Score	Speak_Hours_Score	Use_French_Score	French_Hours_Score	
0	0	0	0	0
Read_French_Score	Write_French_Score	Listen_French_Score	Speak_French_Score	
0	0	0	0	0
Competence_Score	Intention_Score			
0	0			

The Screen_Time variable has 31 missing values. If we were to create a model using Screen_Time we would have to remove all of these observations, despite the fact that they are not missing in the other variables because a linear regression model cannot handle missing values.

Basic Summary Statistics for continuous variables

```
continuous_variables <- c("Class_Level", "Screen_Time", "Interaction_Score",
  "Speaking_Score", "Speak_Hours_Score",
  "Use_French_Score", "French_Hours_Score",
  "Competence_Score", "Intention_Score")
summary(french_data[,continuous_variables])
```

Class_Level	Screen_Time	Interaction_Score	Speaking_Score
Min. :0.000	Min. : 142.5	Min. :13.00	Min. : 0.00
1st Qu.:1.000	1st Qu.: 499.7	1st Qu.:20.25	1st Qu.:14.00
Median :1.000	Median : 915.5	Median :23.00	Median :21.50
Mean :1.552	Mean : 986.6	Mean :23.72	Mean :21.24
3rd Qu.:3.000	3rd Qu.:1128.5	3rd Qu.:27.75	3rd Qu.:27.75
Max. :4.000	Max. :2869.0	Max. :34.00	Max. :35.00
	NA's :31		

Speak_Hours_Score	Use_French_Score	French_Hours_Score	Competence_Score
Min. : 0.000	Min. : 3.00	Min. : 0.000	Min. : 3.00
1st Qu.: 3.000	1st Qu.:11.00	1st Qu.: 1.250	1st Qu.:36.25
Median : 4.500	Median :15.50	Median : 4.000	Median :46.50
Mean : 7.034	Mean :16.17	Mean : 6.103	Mean :46.12
3rd Qu.: 9.500	3rd Qu.:20.00	3rd Qu.: 9.000	3rd Qu.:60.75
Max. :25.000	Max. :28.00	Max. :20.000	Max. :80.00

Intention_Score

Min. : 0.00

1st Qu.: 4.00

Median : 8.50

Mean : 8.31

```
3rd Qu.:11.75
Max.    :20.00
```

It's important to note that Interaction_Score ranged from 0 to 40, Speaking_Score from 0 to 35, Speak_Hours_Score from 0 to 25, Use_French_Score from 0 to 28, French_Hours_Score from 0 to 20, Competence_Score from 0 to 80, and Intention_Score from 0 to 20. So all of these summary statistics look to be within their theoretical ranges. Screen_Time has a very large range from the minimum to the maximum, with the lowest being 2.4 hours per week on average, and the highest being about 47.8 hours per week on average. So that may require more looking into.

Counts for binary variables

```
binary_variables <- c("Year_2024", "Year_2025", "Host_Family", "Phone_Language",
                      "Phone_Keyboard")
```

```
french_data |>
  group_by(Year_2024) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
  Year_2024 count
    <dbl> <int>
1         0    44
2         1    14
```

```
french_data |>
  group_by(Year_2025) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
  Year_2025 count
    <dbl> <int>
1         0    39
2         1    19
```

```
french_data |>
  group_by(Host_Family) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
  Host_Family count
    <dbl> <int>
1         0    35
2         1    23
```

```
french_data |>
  group_by(Phone_Language) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
  Phone_Language count
    <dbl> <int>
1         0    31
2         1    27
```

```
french_data |>
  group_by(Phone_Keyboard) |>
  summarize(count = n())
```

```
# A tibble: 2 x 2
  Phone_Keyboard count
    <dbl> <int>
1         0     9
2         1    49
```

Year_2024, Year_2025, Host_Family, and Phone_Language all have more 0 (no) entries than 1, whereas Phone_Keyboard has more 1s. Phone_Language is the only one that appears to have a roughly equal split, whereas the others are skewed.

Convert binary variables to factor.

```
# Save original numeric data for calculating correlations
french_data_corr <- french_data

# Convert binary variables to factor
french_data <- french_data |>
  mutate(across(all_of(binary_variables), as.factor))
```

Plots

- Color coded by response
 - Default is counts
 - Greens is by Competence Score
 - Blues is by Intention Score

Year

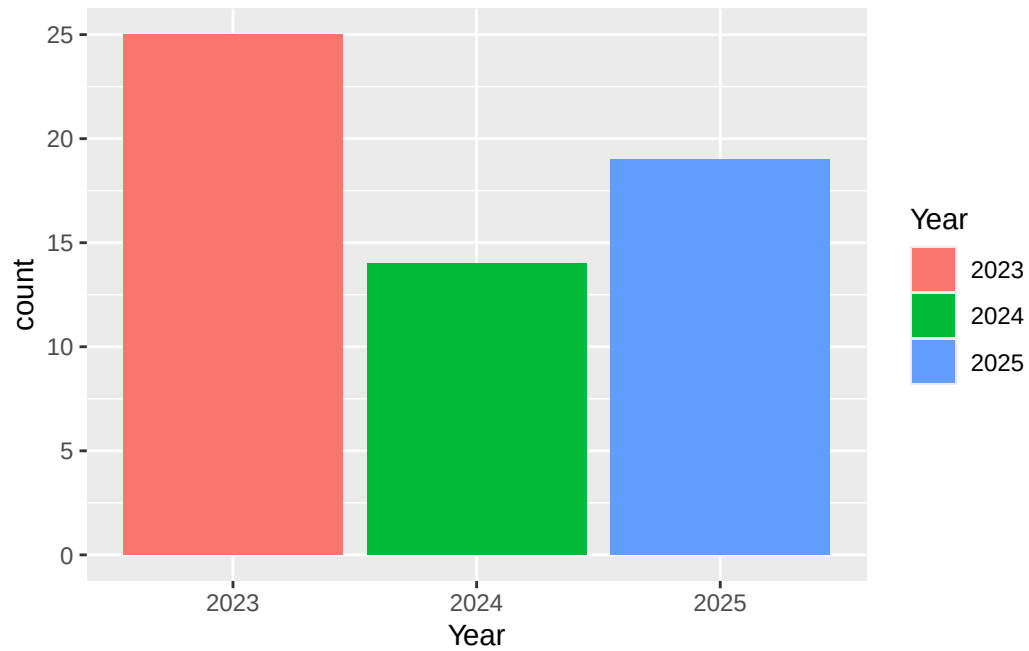
```
french_data <- french_data |>
  mutate(Year = ifelse(Year_2024 == 1, "2024",
                      ifelse(Year_2025 == 1, "2025", "2023")))

french_data |>
  group_by(Year)|>
  summarize(count = n())
```

```
# A tibble: 3 x 2
  Year  count
<chr> <int>
1 2023     25
2 2024     14
3 2025     19
```

Year counts

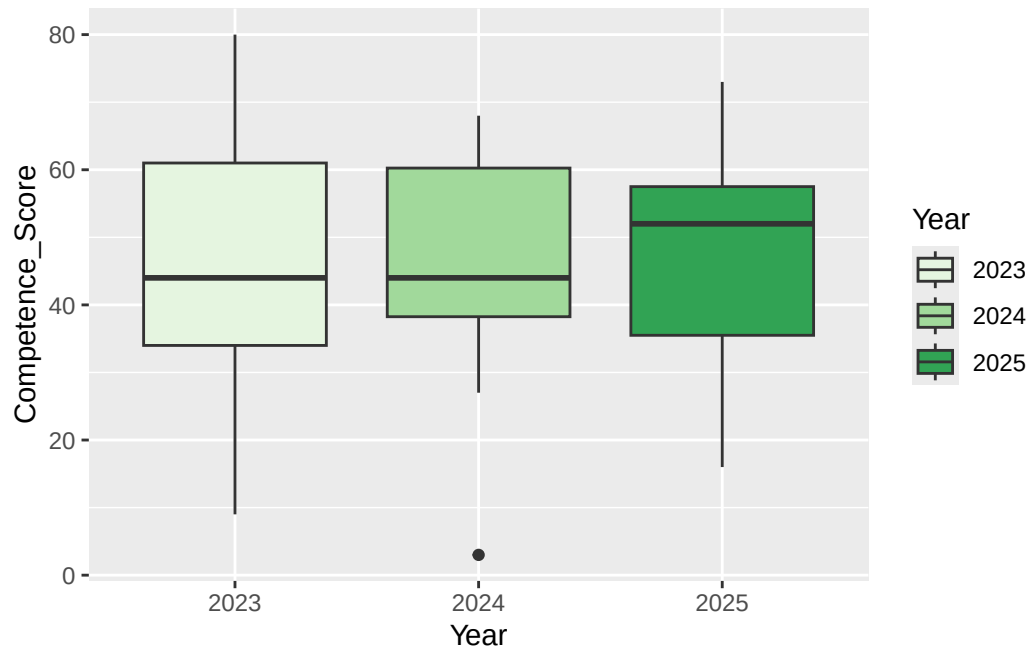
```
ggplot(french_data, aes(Year, fill = Year)) +
  geom_bar()
```



There's a difference in the number of students per year, but no clear pattern. 2023 has the most, then 2025, and finally 2024.

Competence Score by Year

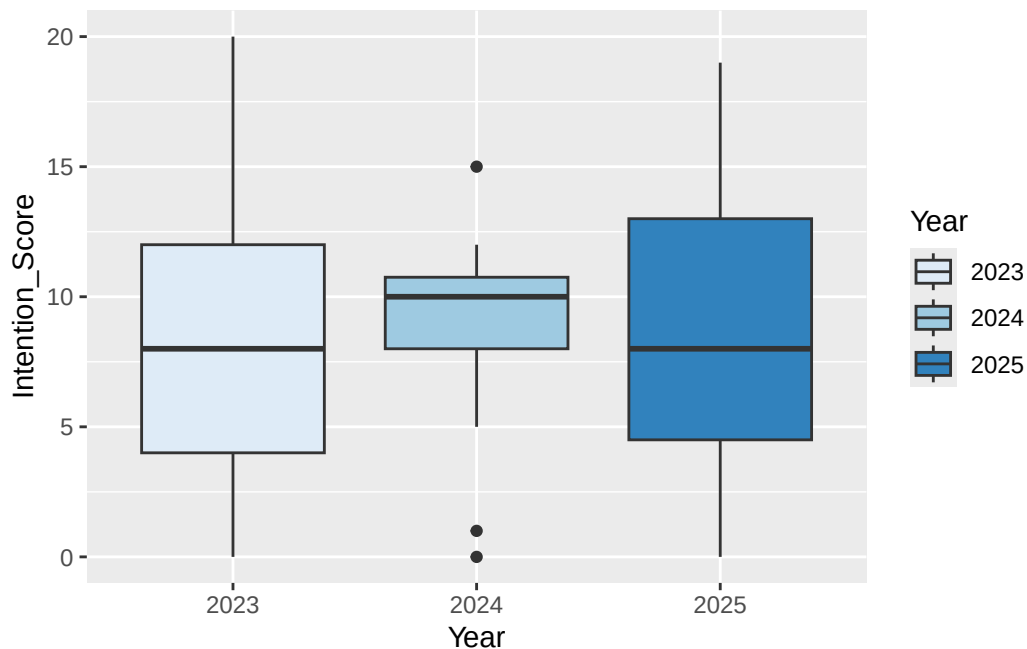
```
ggplot(french_data, aes(Year, Competence_Score, fill = Year)) +  
  geom_boxplot() + scale_fill_brewer(palette = "Greens")
```



For Year vs Competence_Score, there doesn't seem to be any major difference in the overall distributions of the three years. The year 2025 has a higher median Competence Score than 2023 or 2024. 2023 had the widest spread.

Intention Score by Year

```
ggplot(french_data, aes(Year, Intention_Score, fill = Year)) +  
  geom_boxplot() + scale_fill_brewer(palette = "Blues")
```



For Year vs Intention_Score, 2024 has a smaller IQR and range than 2023 and 2025, as well as a slightly higher median Intention_Score. Both 2023 and 2025 have very similar spreads.

Class Level

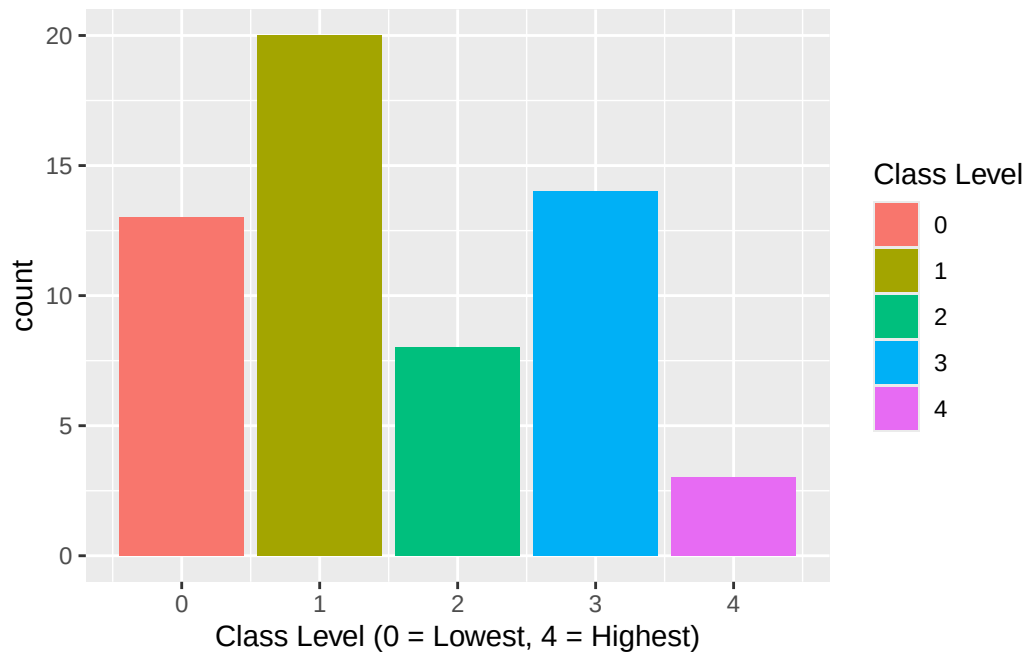
```
french_data |>
  group_by(Class_Level) |>
  summarise(count=n(), avgcompscore = mean(Competence_Score),
            avgintscore = mean(Intention_Score), .groups = "drop")
```

```
# A tibble: 5 x 4
  Class_Level count avgcompscore avgintscore
    <dbl> <int>      <dbl>      <dbl>
1         0    13        32.3         4.15
2         1    20        41.2         7.8
3         2     8        48.4         7.38
4         3    14        59.4        12.9
5         4     3        70.3         11
```

It seems as though by class level, with 4 being the most difficult, competence scores do increase, as one would expect. We do see less of a pattern for intention score.

Class Level counts

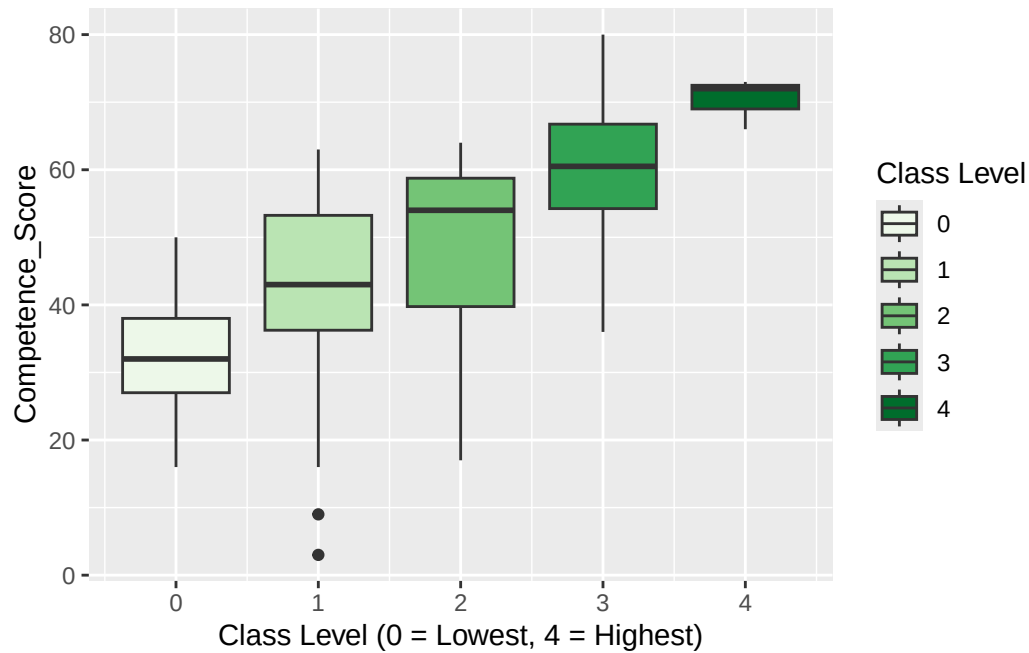
```
ggplot(french_data, aes(Class_Level, fill = factor(Class_Level))) +  
  geom_bar() + labs(x = "Class Level (0 = Lowest, 4 = Highest)", fill="Class Level")
```



The Class_Level is somewhat right skewed with most students in the lowest 2 levels and very few at the highest level.

Competence Score by Class Level

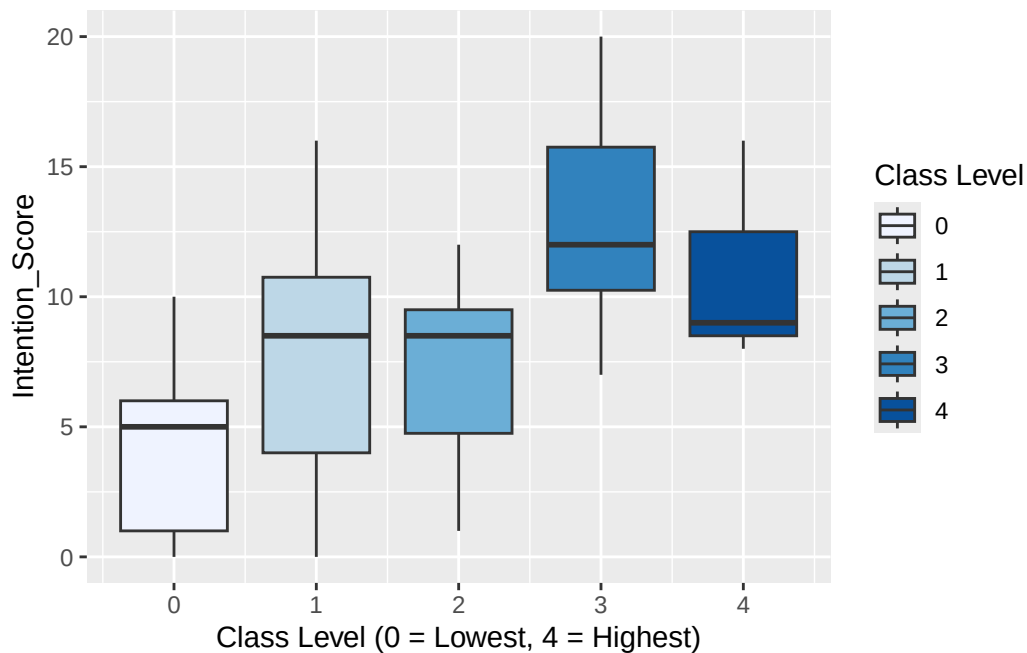
```
ggplot(french_data, aes(Class_Level, Competence_Score,  
  fill = factor(Class_Level))) +  
  geom_boxplot() + scale_fill_brewer(palette = "Greens") +  
  labs(x = "Class Level (0 = Lowest, 4 = Highest)", fill="Class Level")
```



This boxplot shows a clear increase in Competence Score as Class Level increases.

Intention Score by Class Level

```
ggplot(french_data, aes(Class_Level, Intention_Score,
                        fill = factor(Class_Level))) +
  geom_boxplot() + scale_fill_brewer(palette = "Blues") +
  labs(x = "Class Level (0 = Lowest, 4 = Highest)", fill="Class Level")
```



There is less of a clear increase in Intention Score as Class Level increases, we saw in the table before. The individual box plots are also much wider, indicating more variation within the class level categories as well.

Host Family

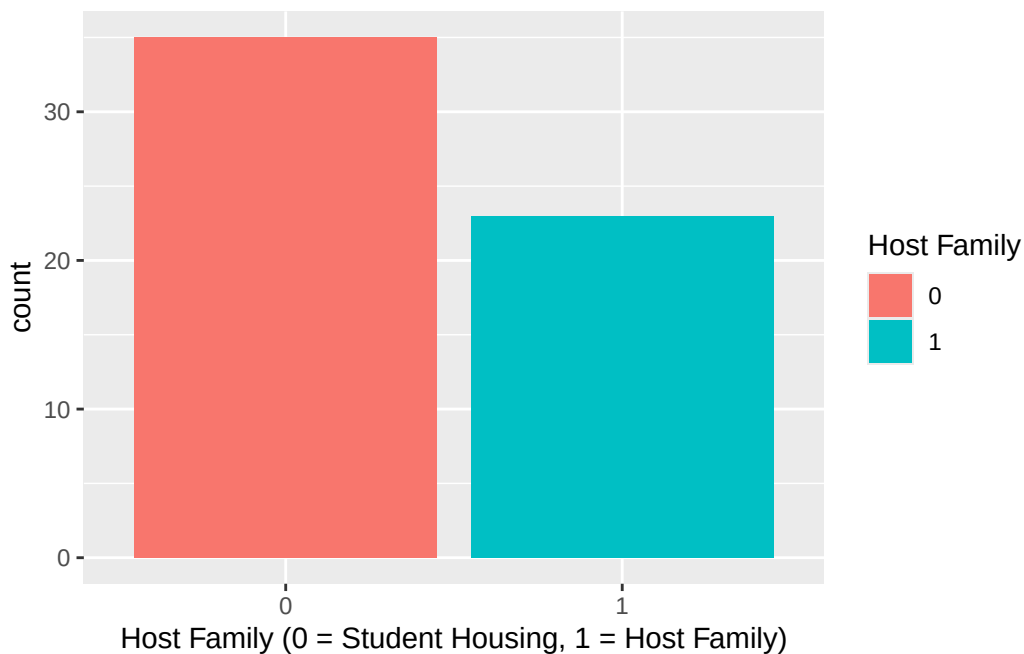
```
french_data |>
  group_by(Host_Family) |>
  summarize(n = n()) |>
  mutate(prop_count = n/sum(n))
```

```
# A tibble: 2 x 3
  Host_Family      n prop_count
  <fct>         <int>     <dbl>
1 0             35     0.603
2 1             23     0.397
```

Top row (Host_Family = 0) is Student housing, and this row seems to be the majority of student living situations.

Host Family counts

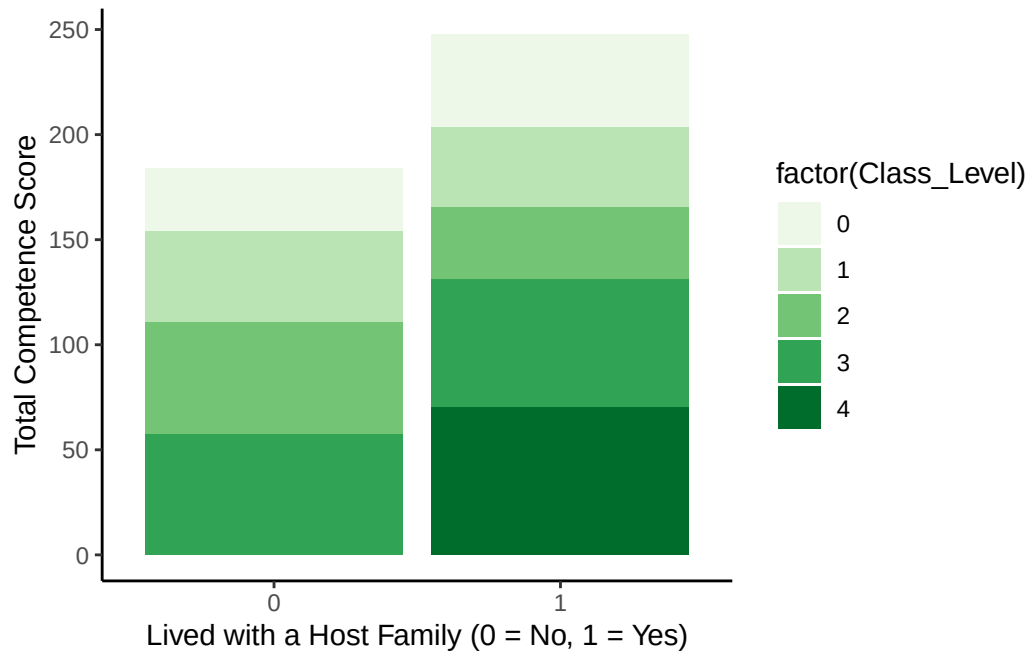
```
ggplot(french_data, aes(Host_Family, group = Host_Family, fill = Host_Family)) +
  geom_bar() + labs(x = "Host Family (0 = Student Housing, 1 = Host Family)", fill = "Host F
```



This graph shows that more students lived in Student housing, visualizing what we saw before in the table

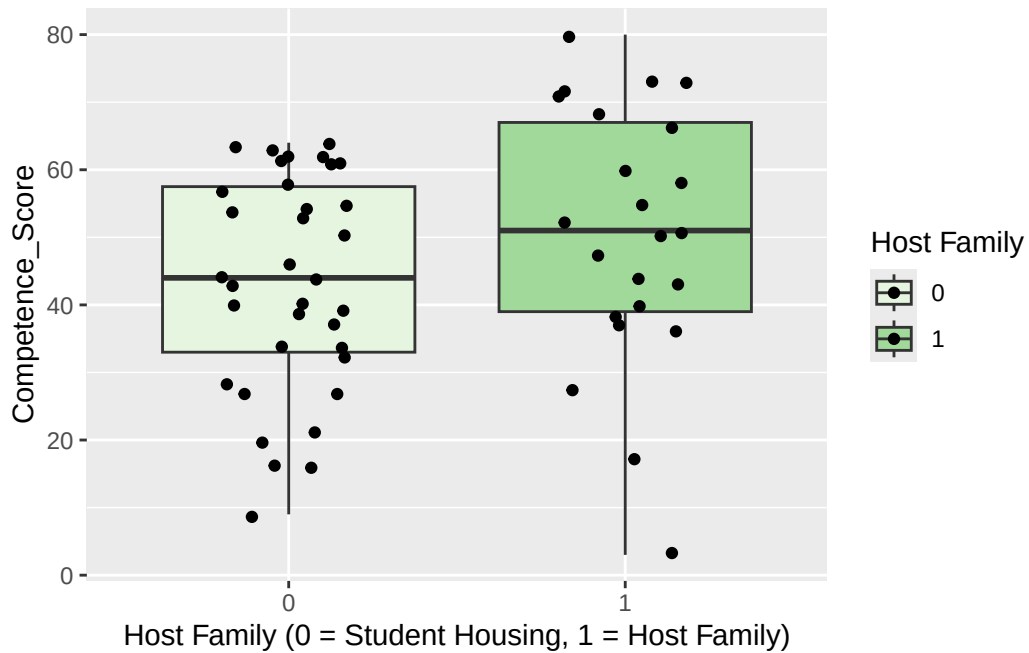
Competence Score by Host Family

```
ggplot(french_data, aes(x = Host_Family, y = Competence_Score,
  fill = factor(Class_Level))) +
  geom_bar(stat = "summary", fun = "mean") +
  scale_fill_brewer(palette = "Greens") +
  labs(x = "Lived with a Host Family (0 = No, 1 = Yes)",
    y = "Total Competence Score") +
  theme_classic()
```



This bar plot shows that those that lived with a Host Family had a higher overall Competence score on average

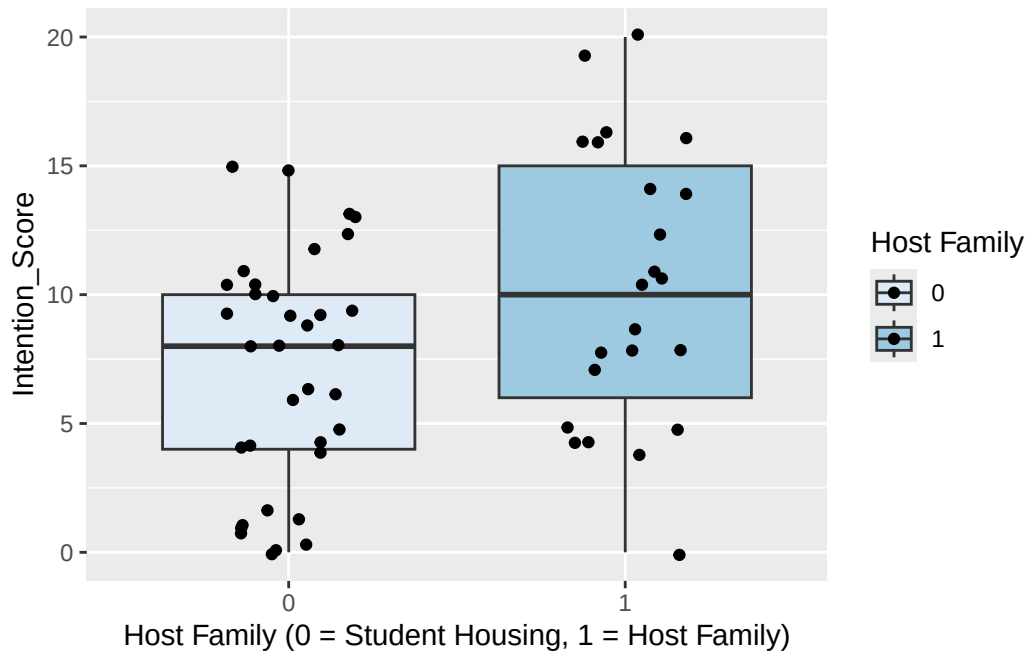
```
ggplot(french_data, aes(Host_Family, Competence_Score, fill = Host_Family)) +
  geom_boxplot() +
  geom_jitter(width=0.2) +
  scale_fill_brewer(palette = "Greens") +
  labs(x = "Host Family (0 = Student Housing, 1 = Host Family)", fill = "Host Family")
```



Overall the students who lived with a host family had a slightly higher competence score than those who lived in student housing.

Intention Score by Host Family

```
ggplot(french_data, aes(Host_Family, Intention_Score, fill = Host_Family)) +
  geom_boxplot() + geom_jitter(width=0.2) +
  scale_fill_brewer(palette = "Blues") +
  labs(x = "Host Family (0 = Student Housing, 1 = Host Family)", fill = "Host Family")
```

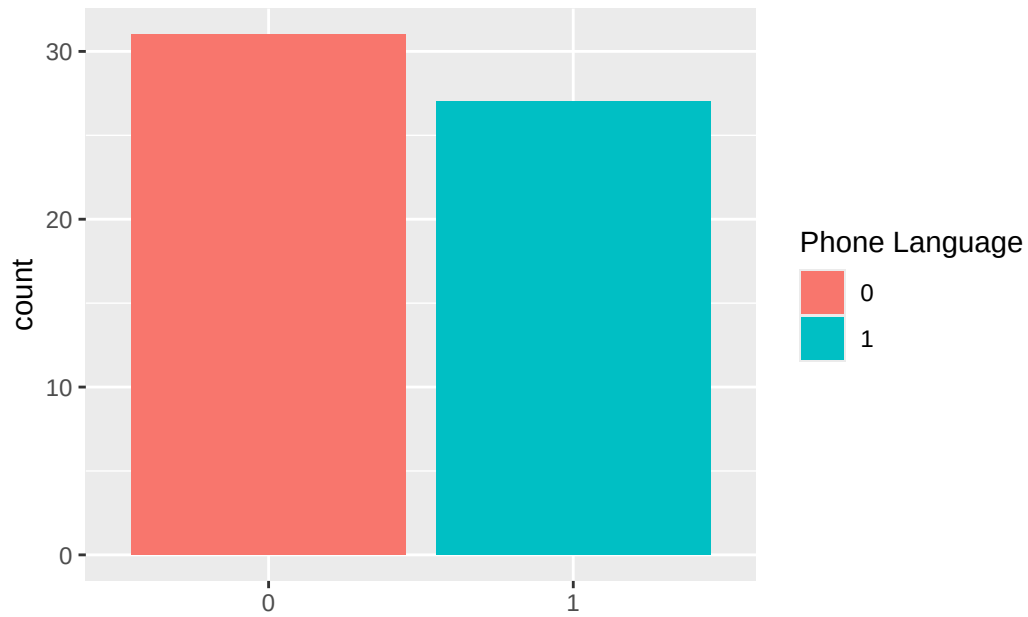


Similarly, overall the students who lived with a host family had a slightly higher intention score. We also see slightly more spread in the Host Family category, and more skew towards higher intention scores in the student housing category

Phone Language

Phone Language counts

```
ggplot(french_data, aes(Phone_Language, fill = Phone_Language)) +
  geom_bar() +
  labs(x = "Phone Language (0 = Did not have French as a Phone Language)",
       fill = "Phone Language")
```

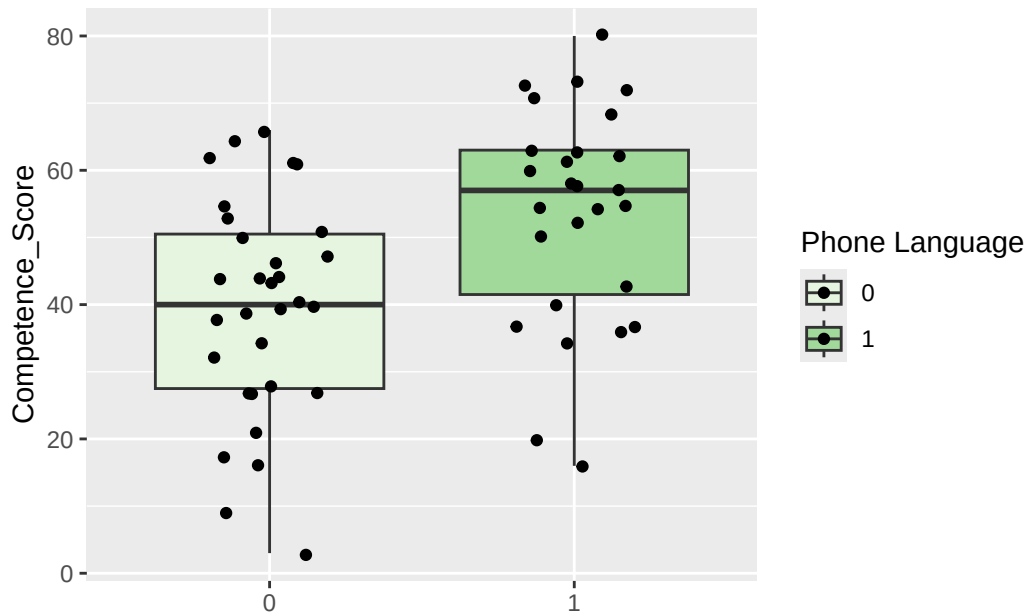


Phone Language (0 = Did not have French as a Phone Language)

Slightly more students did not have French as a language on their phone.

Competence score by Phone Language

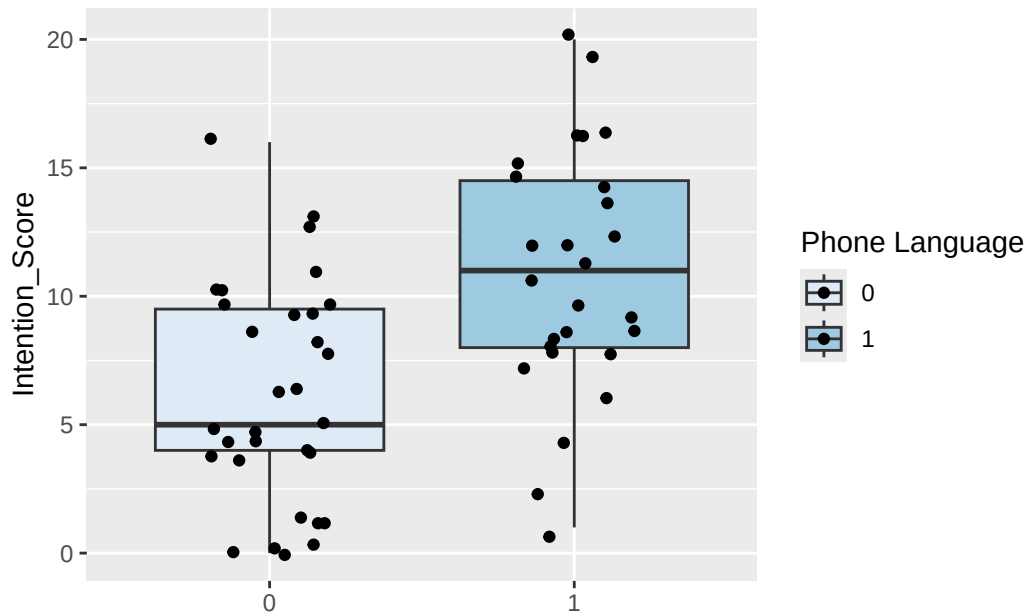
```
ggplot(french_data, aes(Phone_Language, Competence_Score,  
                        fill = Phone_Language)) +  
  geom_boxplot() +  
  geom_jitter(width = 0.2) + scale_fill_brewer(palette = "Greens") +  
  labs(x = "Phone Language (0 = Did not have French as a Phone Language)",  
       fill = "Phone Language")
```



The students who did have French as a language on their phone had a higher median competence score and overall spread.

Intention Score by Phone Language

```
ggplot(french_data, aes(Phone_Language, Intention_Score,
                        fill = Phone_Language)) +
  geom_boxplot() +
  geom_jitter(width = 0.2) + scale_fill_brewer(palette = "Blues") +
  labs(x = "Phone Language (0 = Did not have French as a Phone Language)",
       fill = "Phone Language")
```



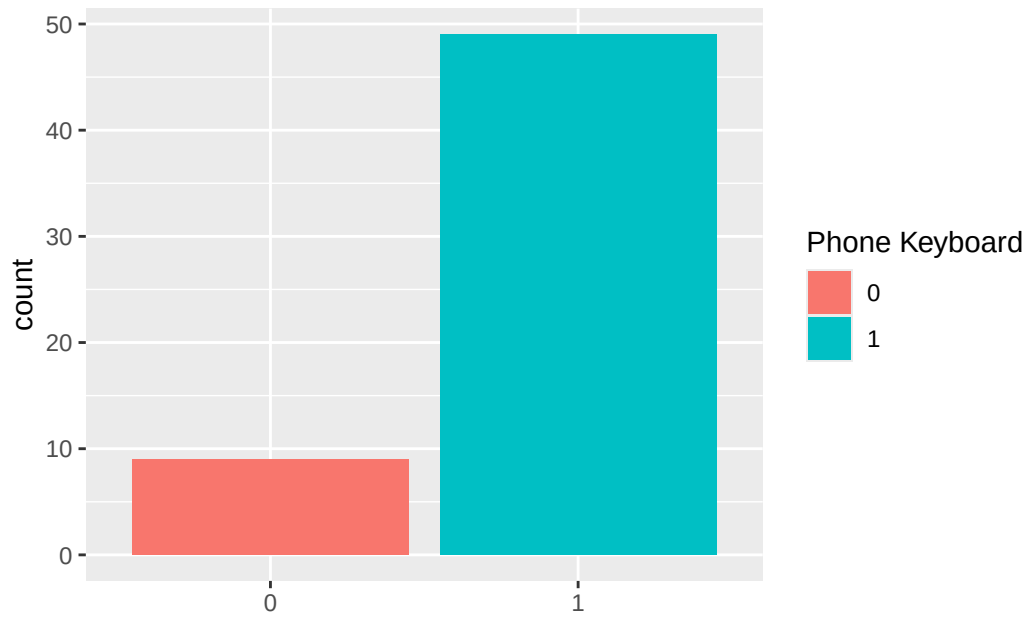
Phone Language (0 = Did not have French as a Phone Language)

Similarly, the students who did have French as a language on their phone had a higher overall spread of intention score.

Phone Keyboard

Phone Keyboard counts

```
ggplot(french_data, aes(Phone_Keyboard, fill = Phone_Keyboard)) +
  geom_bar() +
  labs(x = "Phone Keyboard (0 = Did not have a French Keyboard on their phone)",
       fill = "Phone Keyboard")
```

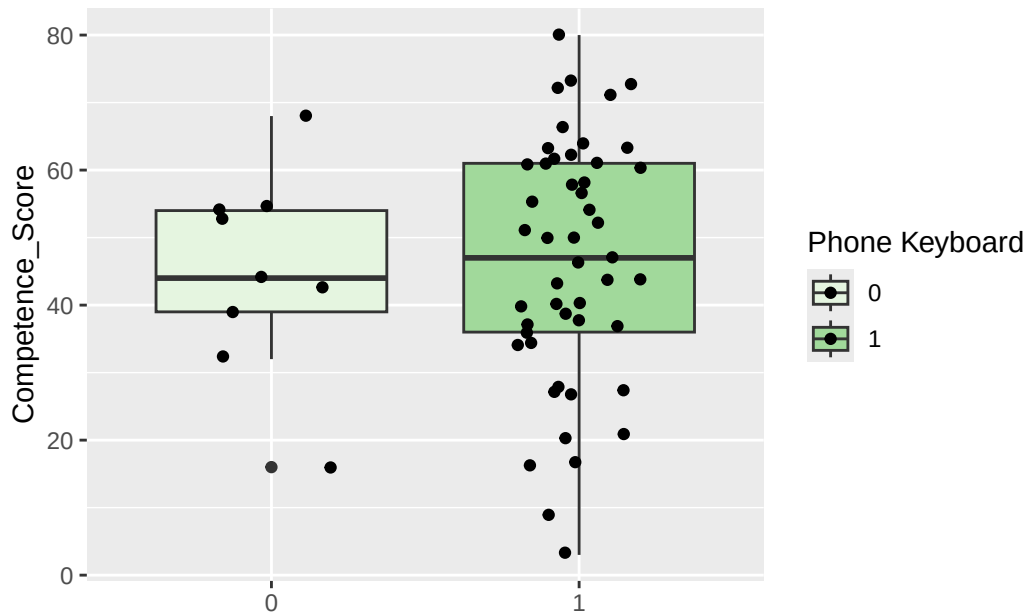


Phone Keyboard (0 = Did not have a French Keyboard on their phone)

There are far more students who have a French keyboard on their phone.

Competence Score by Phone Keyboard

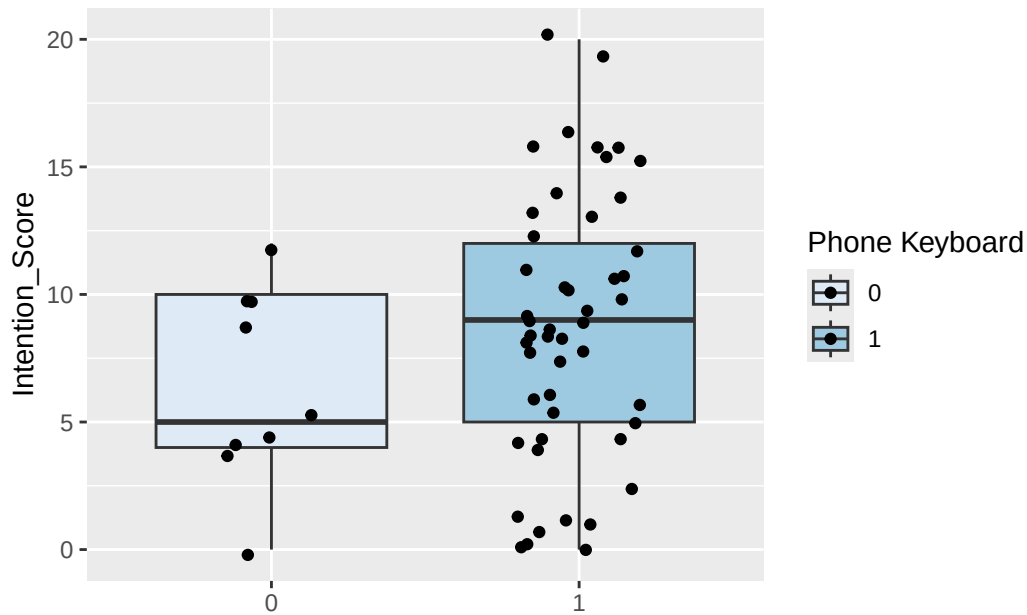
```
ggplot(french_data, aes(Phone_Keyboard, Competence_Score,
                        fill = Phone_Keyboard)) +
  geom_boxplot() + geom_jitter(width = 0.2) + scale_fill_brewer(palette = "Greens") +
  labs(x = "Phone Keyboard (0 = Did not have a French Keyboard on their phone)",
       fill = "Phone Keyboard")
```



While the IQR and range are wider for the French keyboard group as you might expect from having more data, the median competence score for both groups are relatively similar.

Intention Score by Phone Keyboard

```
ggplot(french_data, aes(Phone_Keyboard, Intention_Score,
                        fill = Phone_Keyboard)) +
  geom_boxplot() +
  geom_jitter(width = 0.2) + scale_fill_brewer(palette = "Blues") +
  labs(x = "Phone Keyboard (0 = Did not have a French Keyboard on their phone)",
       fill = "Phone Keyboard")
```

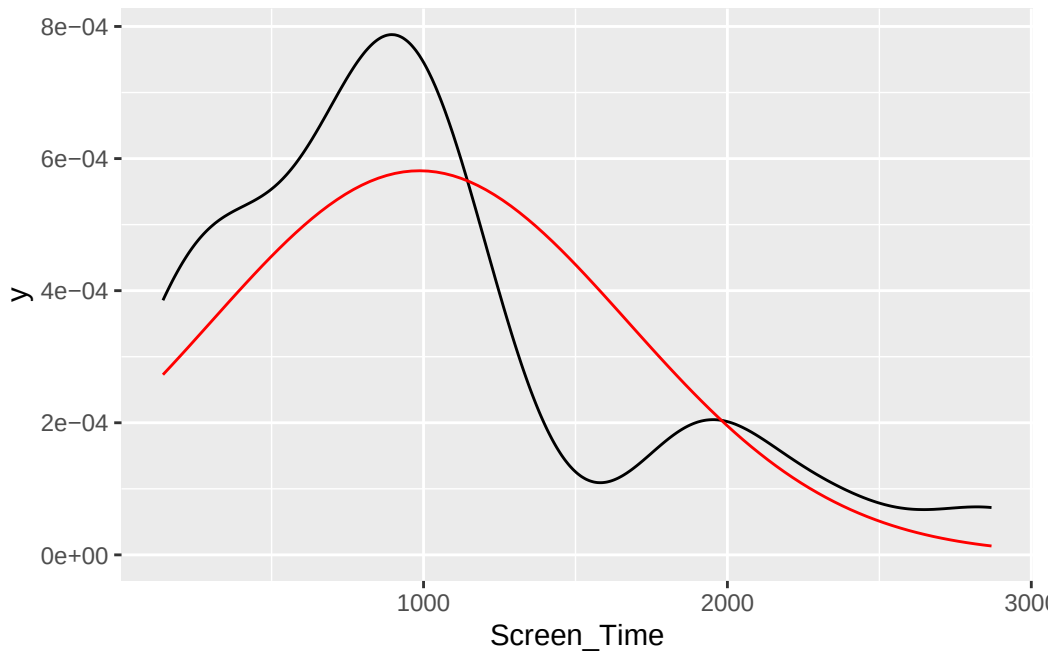


Phone Keyboard (0 = Did not have a French Keyboard on their phone)

The median intention score for the French keyboard group is much higher than the group that didn't use a French keyboard. Also, we see that for the group that did not use a French keyboard, the median is skewed towards the bottom, indicating lower scores.

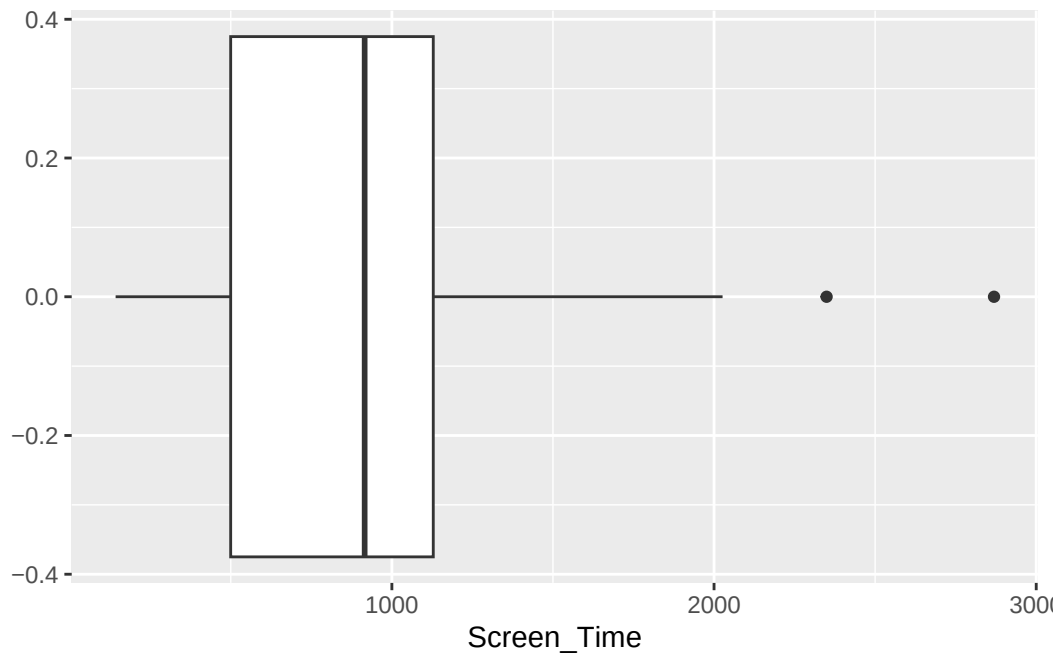
Screen Time

```
# Black line is Screen Time,
# Red line is the Normal dist based off Screen Time
ggplot(french_data, aes(Screen_Time)) +
  geom_density(na.rm = TRUE) +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Screen_Time, na.rm=TRUE),
                             sd = sd(french_data$Screen_Time, na.rm=TRUE)),
               color = "red")
```



It looks like Screen_Time may have a skewed distribution with a higher mean than median. It's also possible that the distribution is bimodal which may indicate that students are answering the question in different ways.

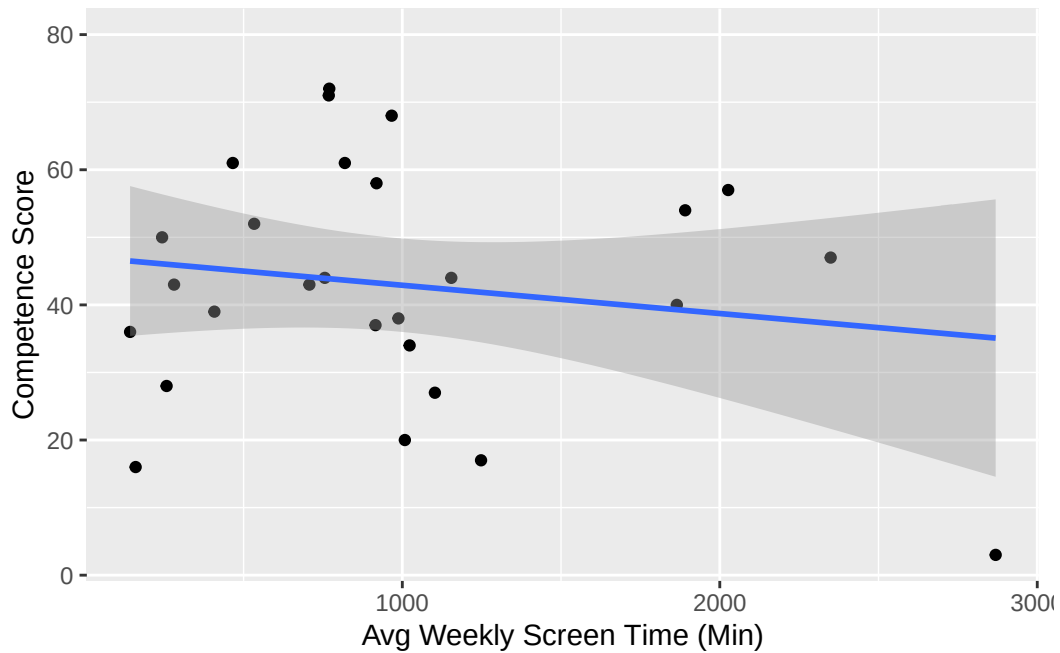
```
ggplot(french_data, aes(x= Screen_Time)) +  
  geom_boxplot(na.rm = TRUE)
```



The boxplot makes it more clear that the reason for the higher mean compared to the median may be due to a few outliers on the far right.

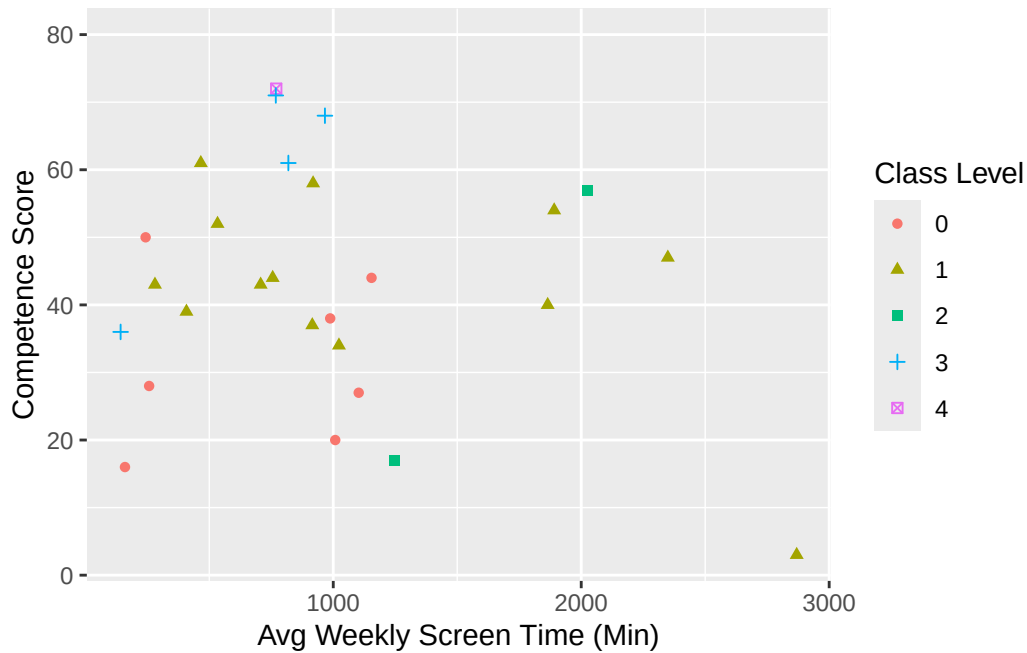
Competence Score by Screen Time

```
ggplot(french_data, aes(Screen_Time, Competence_Score)) +
  geom_point(na.rm=TRUE) +
  geom_smooth(method = 'lm', formula = 'y ~ x', na.rm=TRUE) +
  labs(x = "Avg Weekly Screen Time (Min)", y = "Competence Score")
```



A perfectly flat model is possible in the 95% confidence bands, which indicates there may be no linear relationship between Screen_Time and Competence_Score.

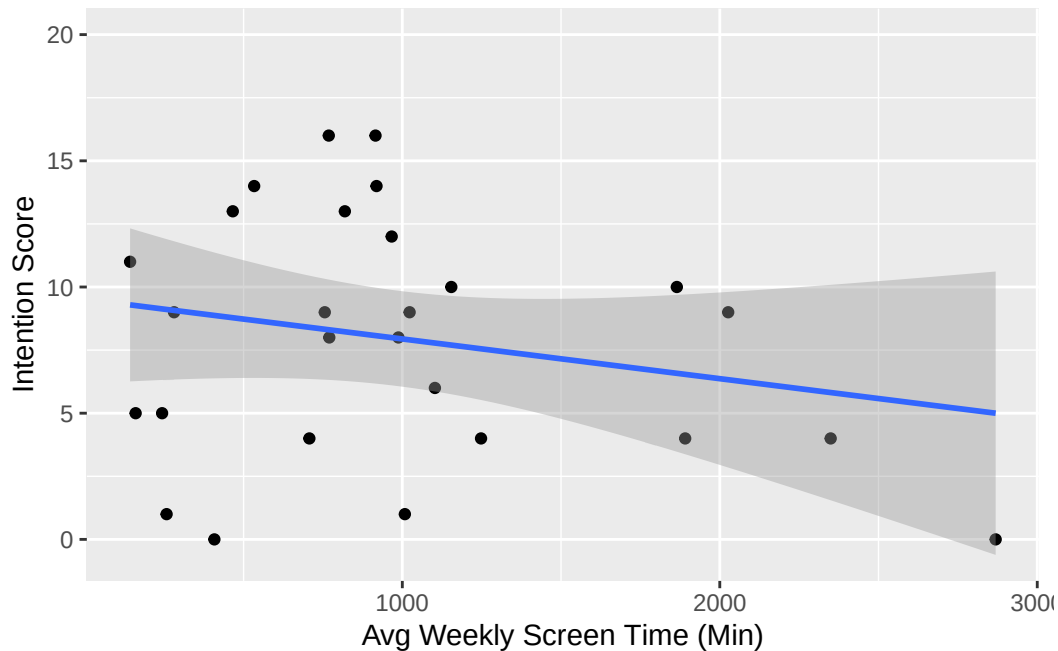
```
ggplot(french_data, aes(x = Screen_Time, y = Competence_Score,
                        color = factor(Class_Level),
                        shape = factor(Class_Level))) +
  geom_point(na.rm = TRUE) +
  labs(x = "Avg Weekly Screen Time (Min)", y = "Competence Score", color = "Class Level", shape = "Class Level")
```



Similar results as those above - no clear linear trend. We do see Class level groupings, with the majority of those on the graph being those from the lower class levels.

Intention Score by Screen Time

```
ggplot(french_data, aes(Screen_Time, Intention_Score)) +
  geom_point(na.rm=TRUE) +
  geom_smooth(method = "lm", formula = 'y ~ x', na.rm=TRUE) +
  labs(x = "Avg Weekly Screen Time (Min)", y = "Intention Score", color = "Class Level", shape = "Class Level")
```

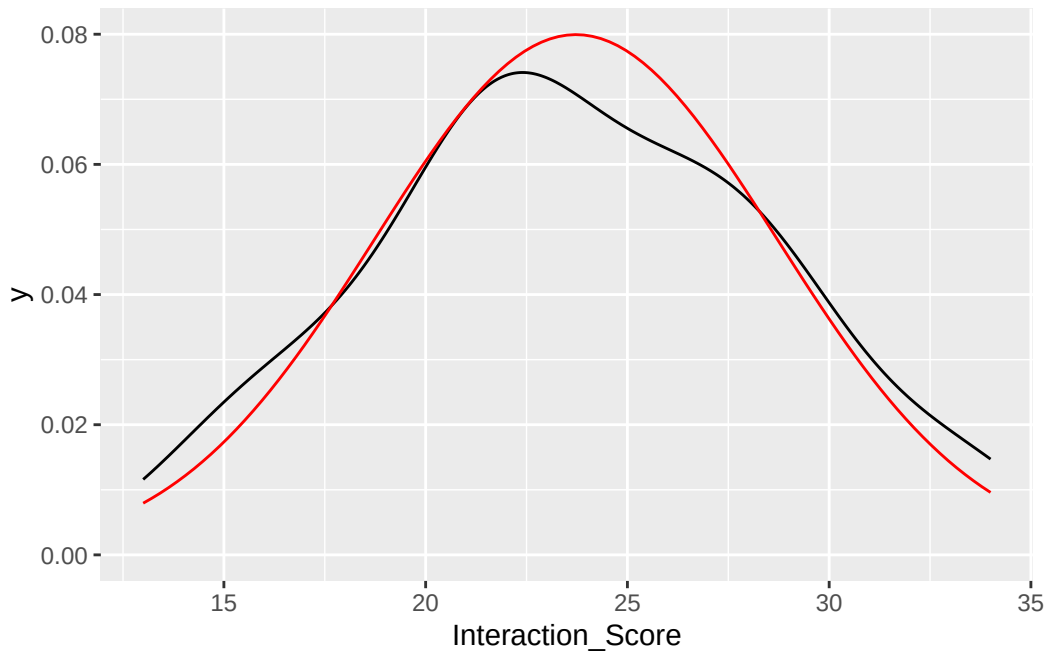


Once again, a flat line is possible in the 95% confidence bands, which indicates that there may not be a linear relationship between Screen_Time and Intention_Score.

Due to the missing values, bimodal nature of the distribution, along with the outliers and the potential for there being no linear relationship with Competence_Score and Intention_Score, Screen_Time will be removed from the model.

Interaction Score

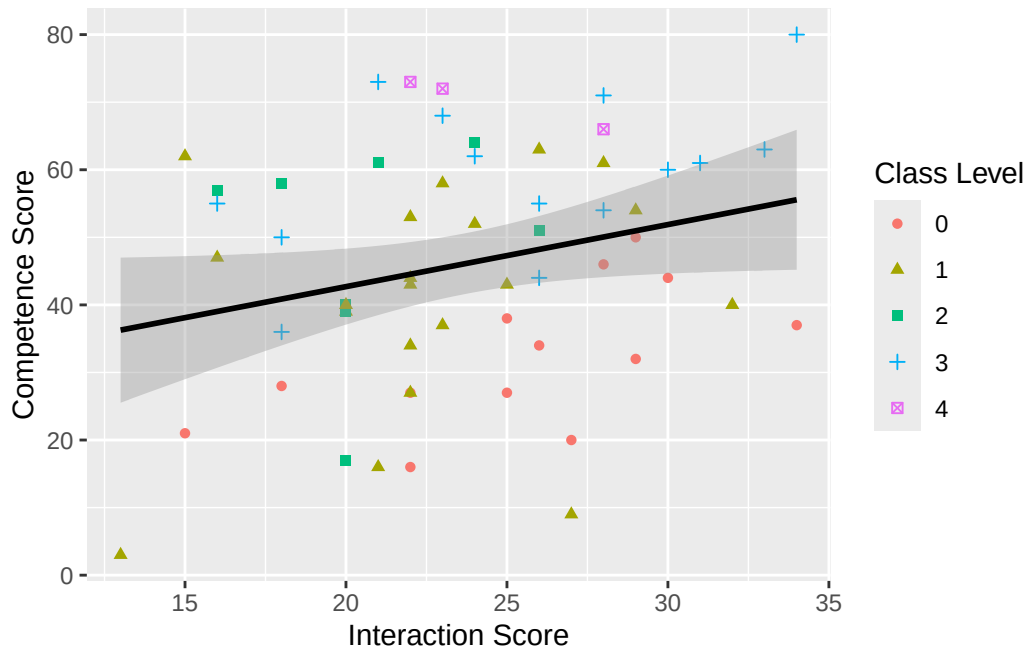
```
# Black line is Interaction Score
# Red line is the Normal dist based off Interaction Score
ggplot(french_data, aes(Interaction_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Interaction_Score),
                           sd = sd(french_data$Interaction_Score)), color = "red")
```



Interaction_Score appears to follow the normal distribution quite well given the amount of data in the sample.

Competence Score by Interaction Score

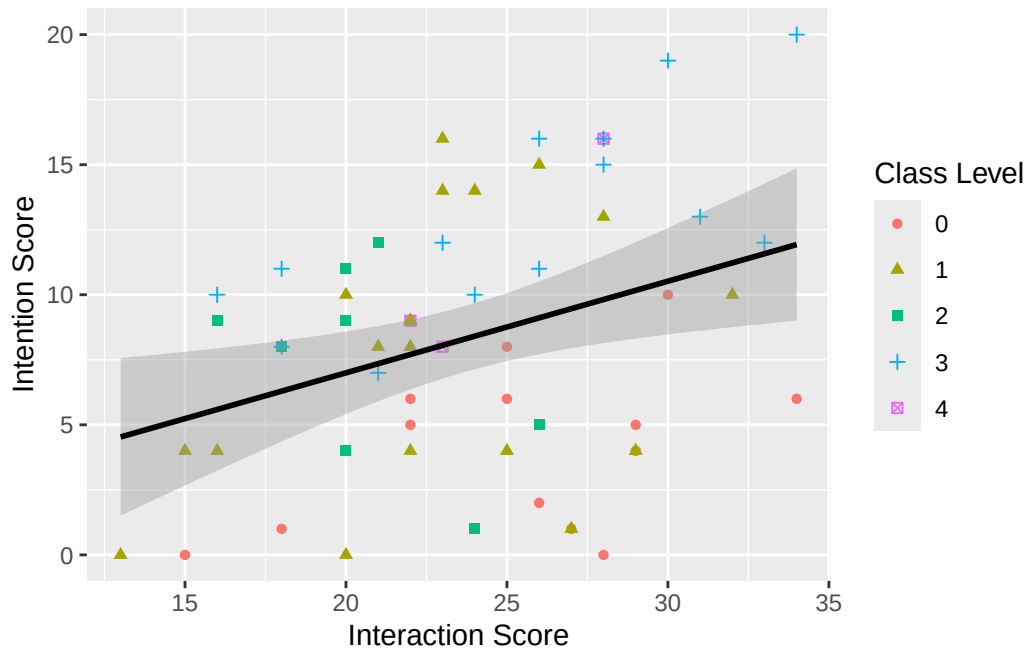
```
ggplot(french_data, aes(Interaction_Score, Competence_Score)) +  
  geom_point(aes(color = factor(Class_Level),  
                 shape = factor(Class_Level))) +  
  geom_smooth(method = 'lm', formula = 'y ~ x', color = "black") +  
  labs(x = "Interaction Score", y = "Competence Score", color = "Class Level",  
       shape = "Class Level")
```



There appears to be a slight positive linear relationship between Interaction_Score and Competence_Score, however a flat line may be possible in the 95% confidence bands, which would indicate no relationship. There are again visible class level groupings, with the lower class levels being more towards the bottom of the graph and the higher levels being more towards the top. Once again we see the class level groupings

Intention Score by Interaction Score

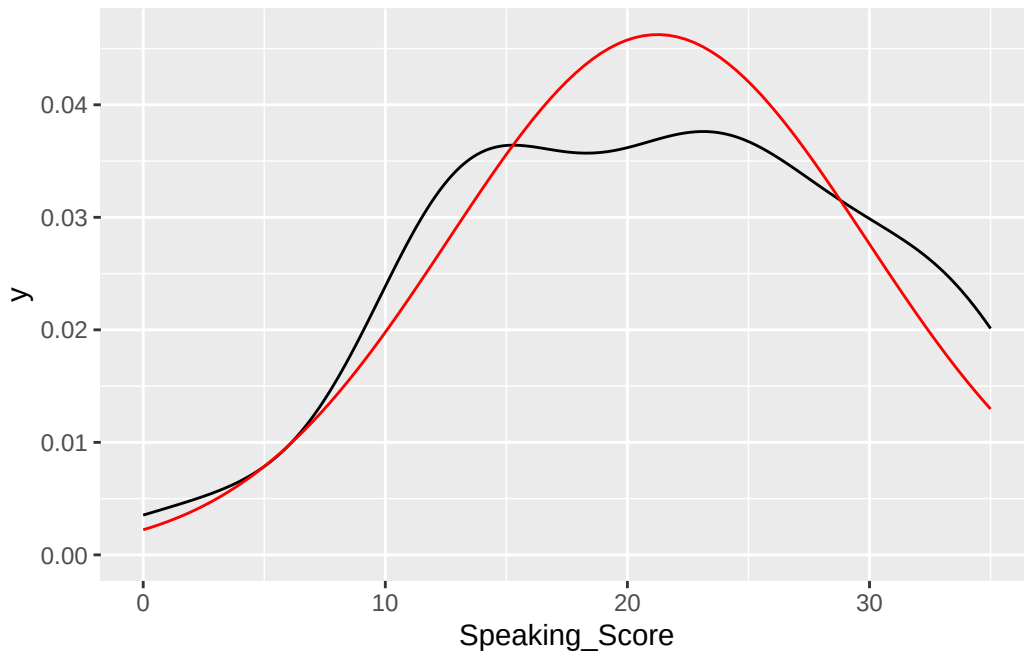
```
ggplot(french_data, aes(Interaction_Score, Intention_Score)) +
  geom_point(aes(color = factor(Class_Level),
                             shape = factor(Class_Level))) +
  geom_smooth(method = 'lm', formula = 'y ~ x', color = "black") +
  labs(x = "Interaction Score", y = "Intention Score", color = "Class Level",
       shape = "Class Level")
```



This plot indicates that there is a positive linear relationship between Interaction_Score and Intention_Score.

Speaking Score

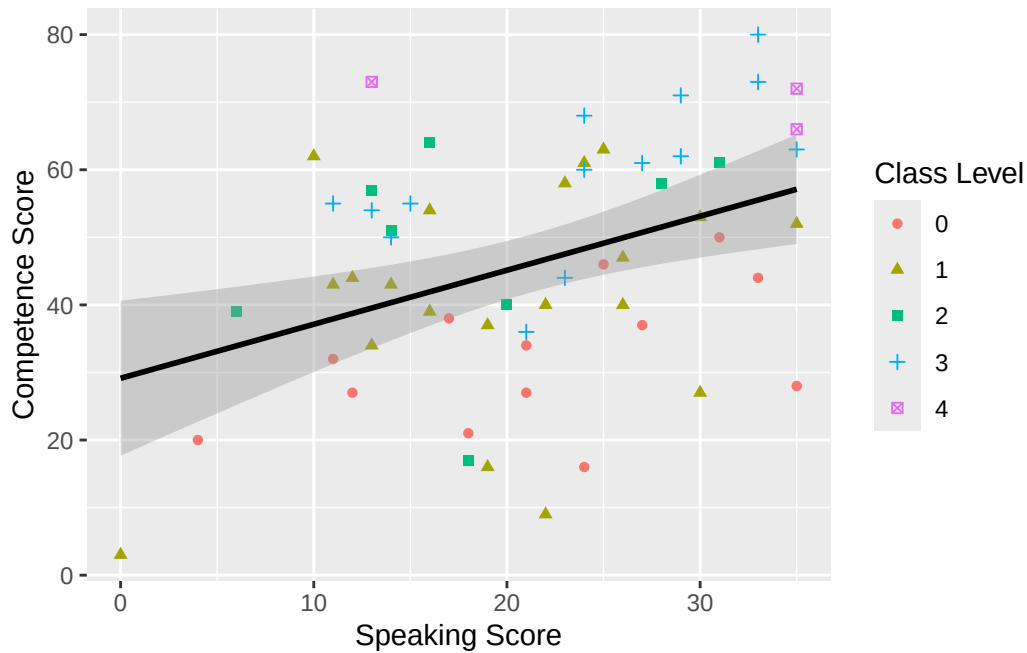
```
# Black line is Speaking Score
# Red line is the Normal dist based off Speaking Score
ggplot(french_data, aes(Speaking_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Speaking_Score),
                           sd = sd(french_data$Speaking_Score)), color = "red")
```



Given the amount of data in the sample, Speaking_Score is roughly following a normal distribution, however it's possible that it deviates slightly. From the summary statistics above, the mean and median are very close to one another.

Competence Score by Speaking Score

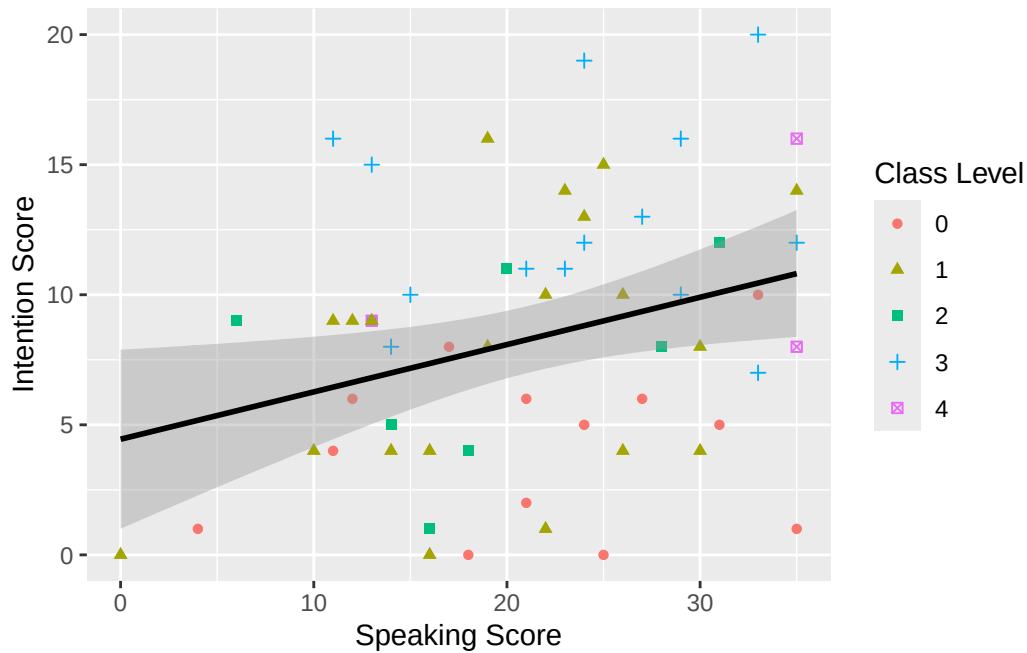
```
ggplot(french_data, aes(Speaking_Score, Competence_Score)) +
  geom_point(aes(color = factor(Class_Level),
    shape = factor(Class_Level))) +
  geom_smooth(method = 'lm', formula = 'y ~ x', color = "black") +
  labs(x = "Speaking Score", y = "Competence Score", color = "Class Level",
    shape = "Class Level")
```



There appears to be a positive linear relationship between Speaking_Score and Competence_Score.

Intention Score by Speaking Score

```
ggplot(french_data, aes(Speaking_Score, Intention_Score)) +
  geom_point(aes(color = factor(Class_Level),
                        shape = factor(Class_Level))) +
  geom_smooth(method = 'lm', formula = 'y ~ x', color = "black") +
  labs(x = "Speaking Score", y = "Intention Score", color = "Class Level",
        shape = "Class Level")
```



There also appears to be a slight positive linear relationship between Speaking_Score and Intention_Score.

Speak Hours Score

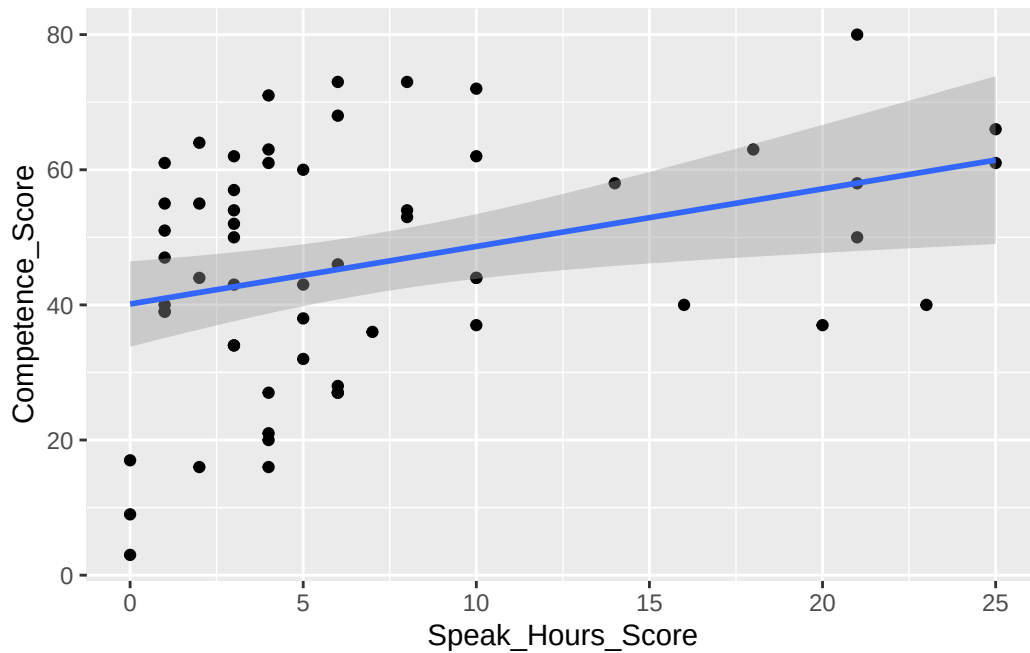
```
# Black line is Speak Hours Score
# Red line is the Normal dist based off Speak Hours Score
ggplot(french_data, aes(Speak_Hours_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Speak_Hours_Score),
                           sd = sd(french_data$Speak_Hours_Score)), color = "red")
```



The `Speak_Hours_Score` appears to be left skewed and from the summary statistics it has a much higher mean than median.

Competence Score by Speak Hours Score

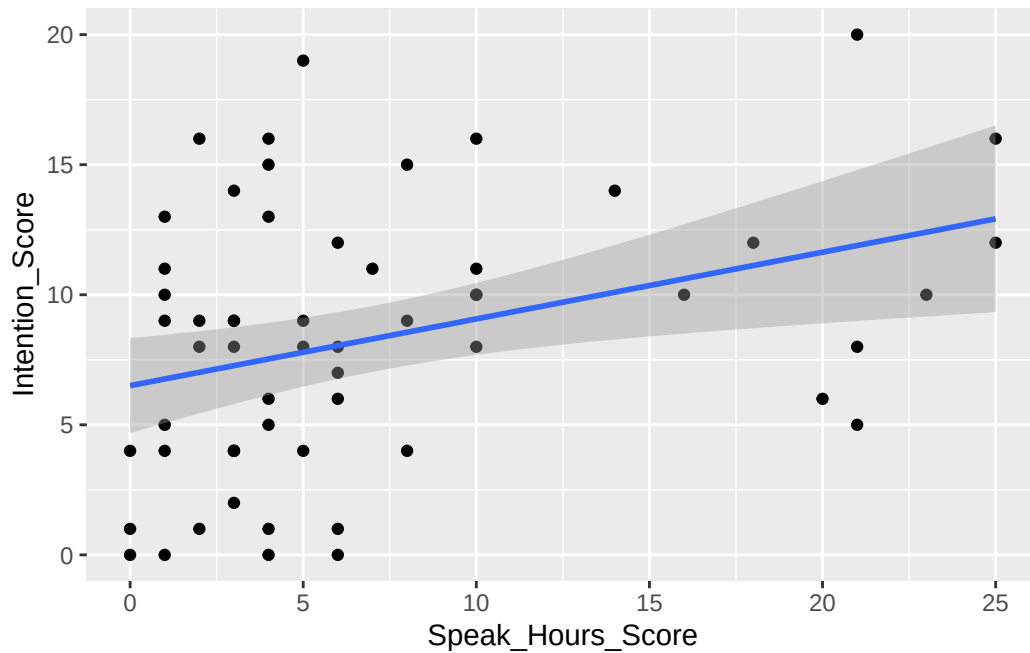
```
ggplot(french_data, aes(Speak_Hours_Score, Competence_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



There is a positive linear relationship between `Speak_Hours_Score` and `Competence_Score`. The points are also much more clustered towards the beginning of the graph.

Intention Score by Speak Hours Score

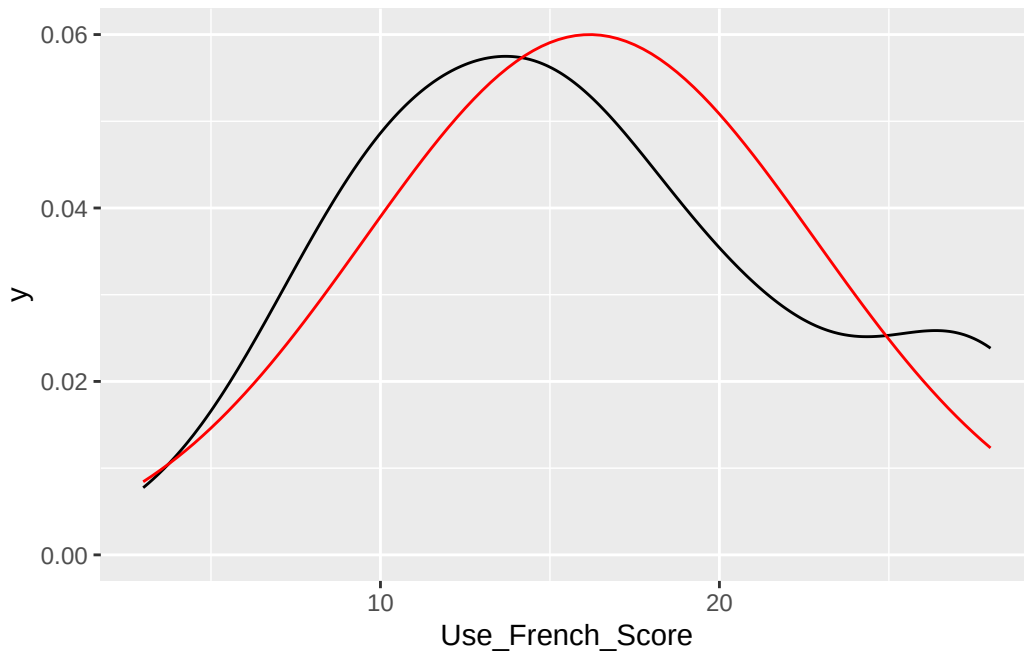
```
ggplot(french_data, aes(Speak_Hours_Score, Intention_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



Similarly, there is a positive linear relationship between `Speak_Hours_Score` and `Intention_Score`.

Use French Score

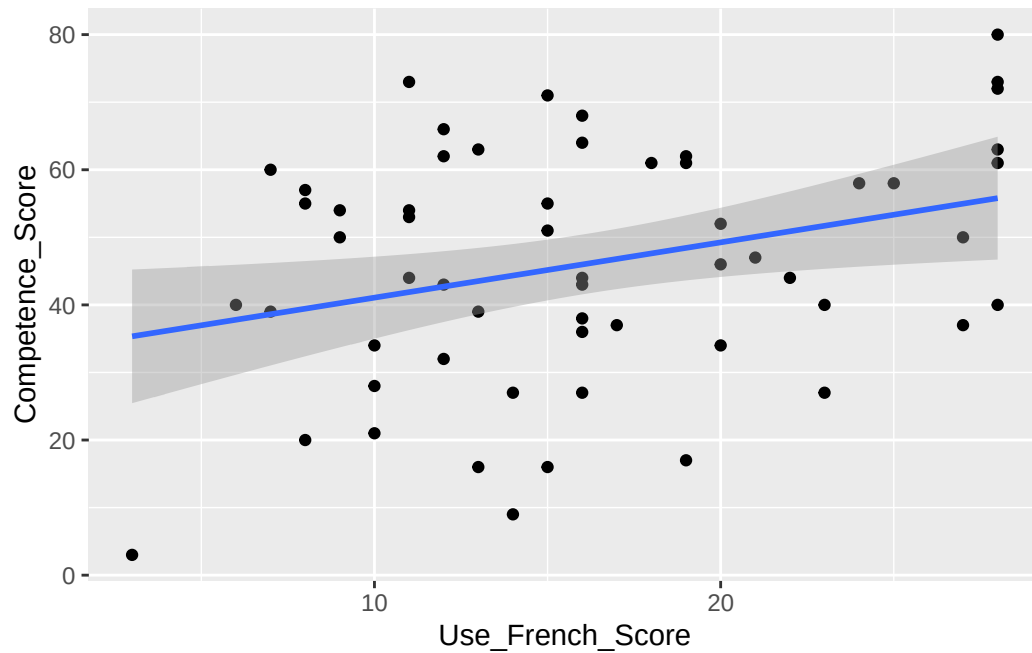
```
# Black line is Use French Score
# Red line is the Normal dist based off Use French Score
ggplot(french_data, aes(Use_French_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Use_French_Score),
                           sd = sd(french_data$Use_French_Score)), color = "red")
```



The density shows that `Use_French_Score` may be slightly left skewed, which is backed up by the summary statistics which show that the mean is higher than the median. However, given the relatively small sample size, it's possible this is just due to variance.

Competence Score by Use French Score

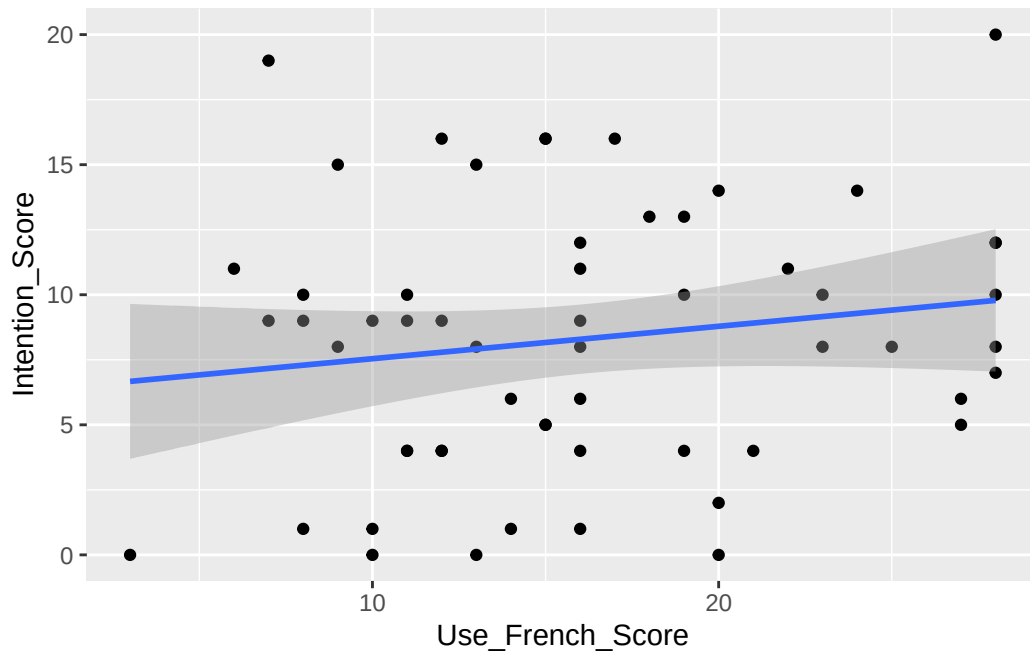
```
ggplot(french_data, aes(Use_French_Score, Competence_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



There is a slight positive linear relationship between Use_French_Score and Competence_Score.

Intention Score by Use French Score

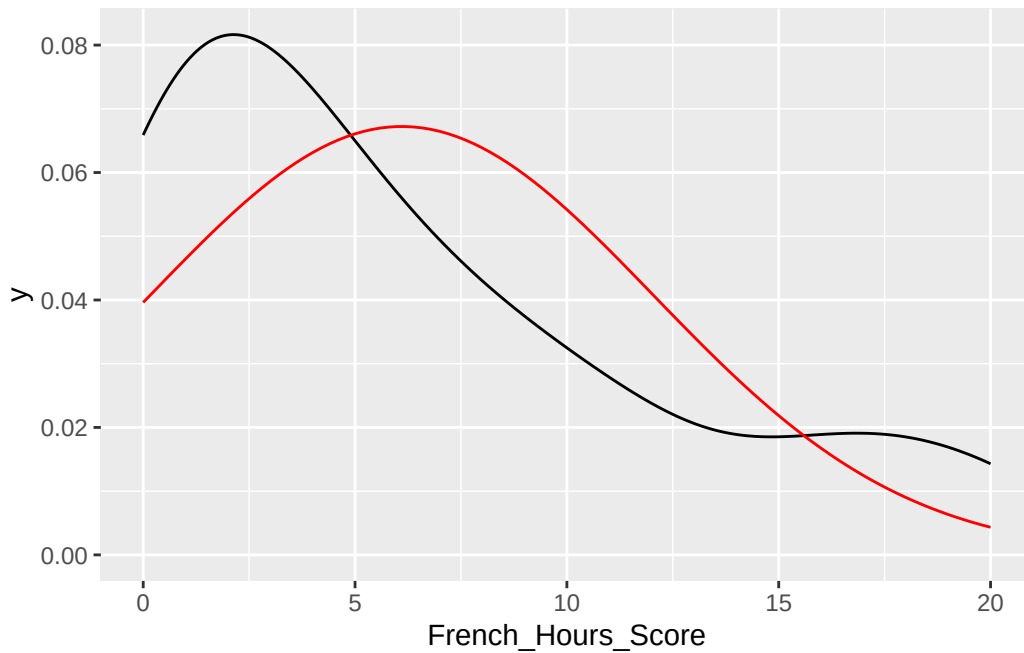
```
ggplot(french_data, aes(Use_French_Score, Intention_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



A flat line between Use_French_Score and Intention_Score is possible in the 95% confidence bands, which indicates there may not be a linear relationship.

French Hours Score

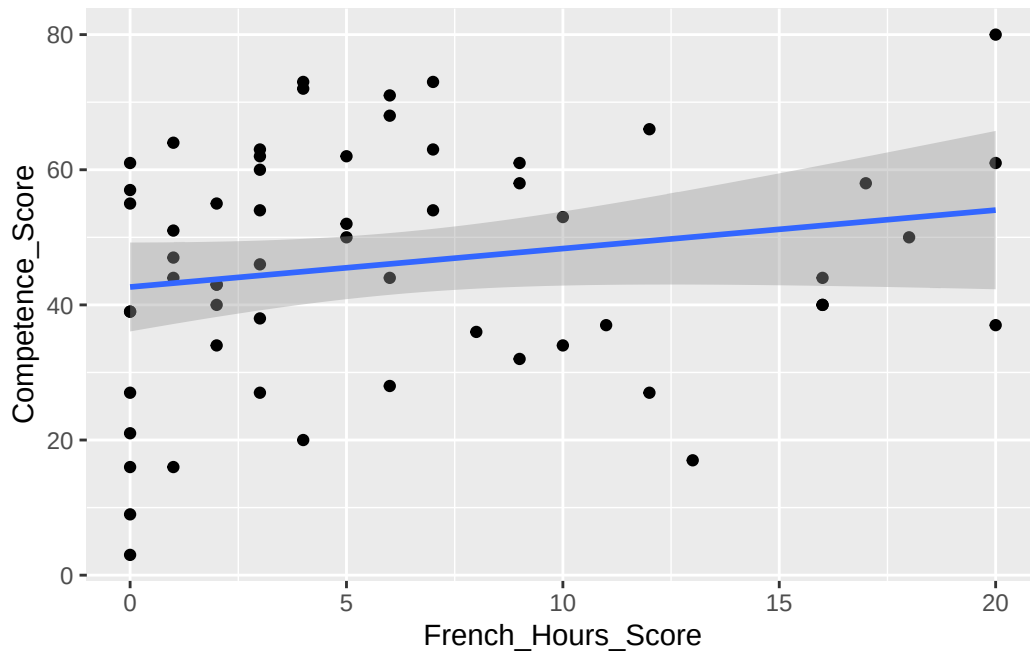
```
# Black line is French Hours Score
# Red line is the Normal dist based off French Hours Score
ggplot(french_data, aes(French_Hours_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$French_Hours_Score),
                           sd = sd(french_data$French_Hours_Score)), color = "red")
```



The French_Hours_Score is left skewed which is also backed up by the summary statistics showing that it has a much higher mean than median.

Competence Score by French Hours Score

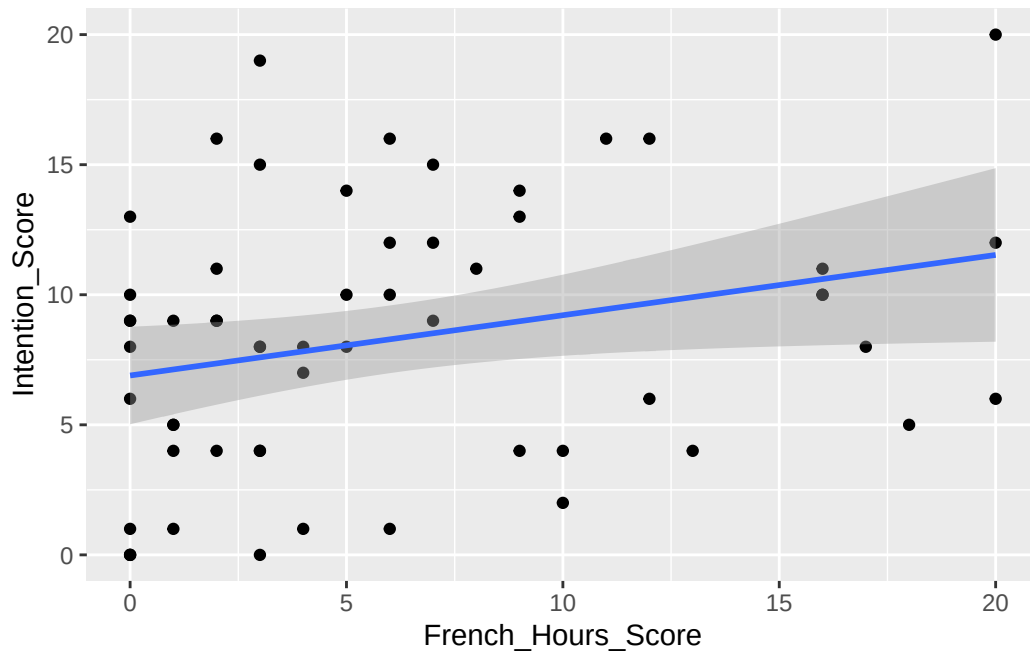
```
ggplot(french_data, aes(French_Hours_Score, Competence_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



A flat line is possible in the 95% confidence bands, therefore there may not be a linear relationship between French_Hours_Score and Competence_Score.

Intention Score by French Hours Score

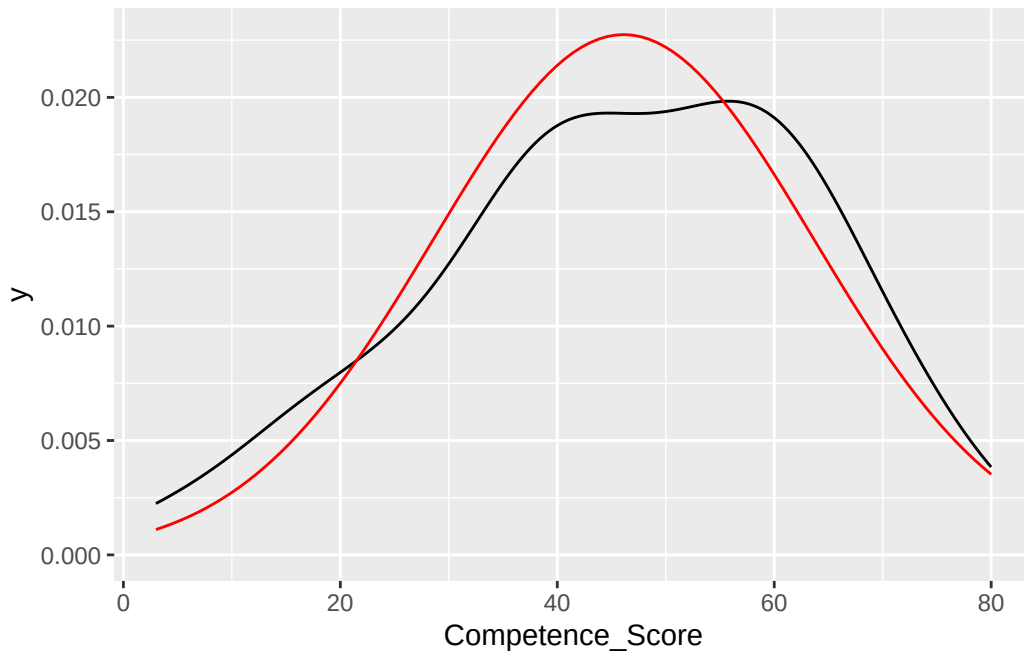
```
ggplot(french_data, aes(French_Hours_Score, Intention_Score)) +
  geom_point() +
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



A flat line is also possible in the 95% confidence bands here, therefore there may not be a linear relationship between French_Hours_Score and Intention_Score.

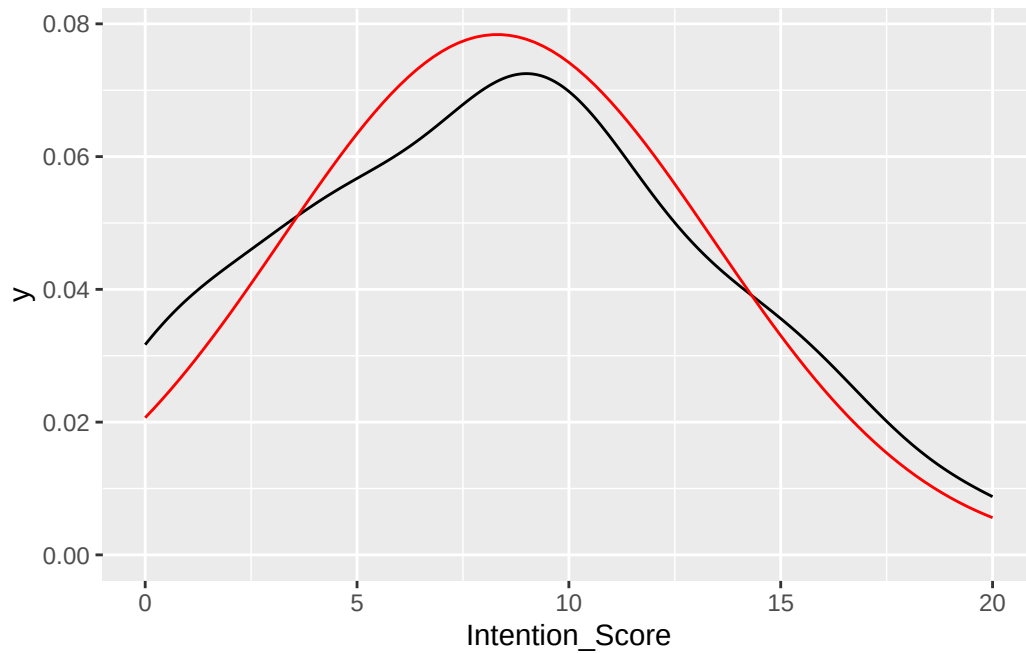
Competence and Intention Score

```
# Black line is Competence Score
# Red line is the Normal dist based off Competence Score
ggplot(french_data, aes(Competence_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Competence_Score),
                           sd = sd(french_data$Competence_Score)), color = "red")
```



Given the sample size, the Competence_Score roughly follows the normal distribution. Additionally, looking at the summary statistics, the mean and median were very close.

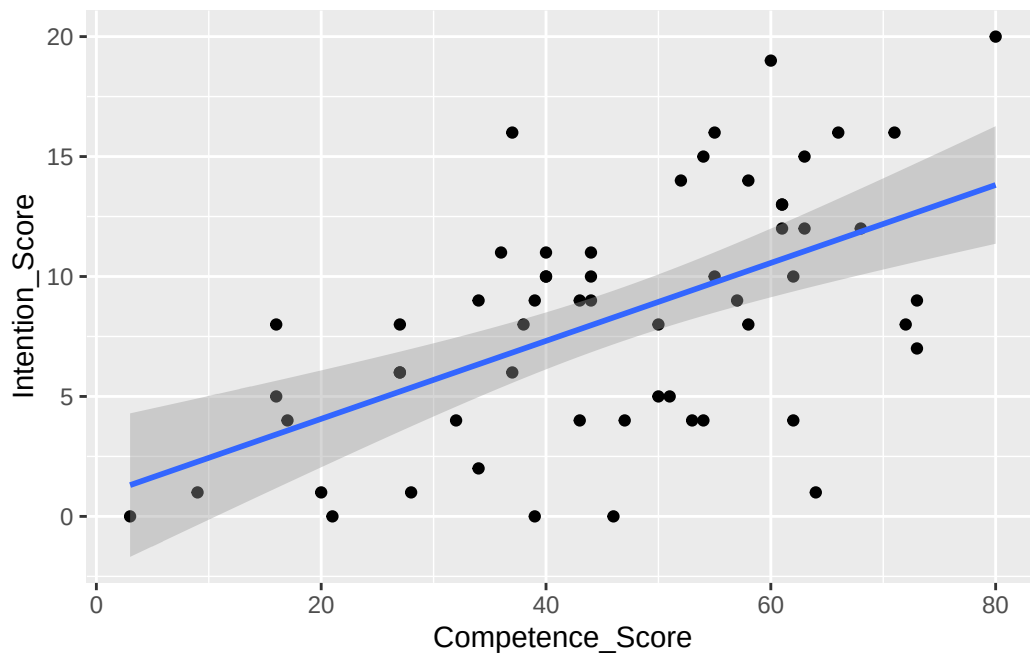
```
# Black line is Intention Score
# Red line is the Normal dist based off Intention Score
ggplot(french_data, aes(Intention_Score)) +
  geom_density() +
  stat_function(fun = dnorm,
               args = list(mean = mean(french_data$Intention_Score),
                           sd = sd(french_data$Intention_Score)), color = "red")
```



Similarly, given the sample size, the Intention_Score also roughly follows a normal distribution. It's mean and median were also very close to one another.

Intention Score by Competence Score

```
ggplot(french_data, aes(Competence_Score, Intention_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



There is a very clear positive linear relationship between Competence_Score and Intention_Score.

Correlations

```
# Look at the correlations with Screen_Time separately since it removes NA values
ST_corr <- french_data_corr |>
  dplyr::select(Year_2024, Year_2025, Class_Level, Host_Family, Phone_Language,
    Phone_Keyboard, Screen_Time, Interaction_Score, Speaking_Score,
    Speak_Hours_Score, Use_French_Score, French_Hours_Score,
    Competence_Score, Intention_Score) |>
  cor(use = "complete.obs")

ST_corr[, "Screen_Time"]
```

Year_2024	Year_2025	Class_Level	Host_Family
0.168152868	-0.250053741	0.007219398	0.065436853
Phone_Language	Phone_Keyboard	Screen_Time	Interaction_Score
-0.070620580	0.140079375	1.000000000	-0.244909593
Speaking_Score	Speak_Hours_Score	Use_French_Score	French_Hours_Score
-0.370320742	-0.129885693	-0.231188348	-0.174212621

Competence_Score	Intention_Score
-0.165508975	-0.224974023

As can be seen, the Screen_Time variable has low correlations with all of the other variables, including weak correlations with both Competence_Score and Intention_Score. This is more evidence that Screen_Time may need to be removed from the data set.

```
# Correlations for the rest of the data
french_data_corr |>
  dplyr::select(Year_2024, Year_2025, Class_Level, Host_Family, Phone_Language,
    Phone_Keyboard, Interaction_Score, Speaking_Score,
    Speak_Hours_Score, Use_French_Score, French_Hours_Score,
    Competence_Score, Intention_Score) |>
  cor(use = "complete.obs")
```

	Year_2024	Year_2025	Class_Level	Host_Family
Year_2024	1.00000000	-0.39371520	0.10817604	0.201650846
Year_2025	-0.39371520	1.00000000	-0.01453506	-0.190331334
Class_Level	0.10817604	-0.01453506	1.00000000	0.355605207
Host_Family	0.20165085	-0.19033133	0.35560521	1.000000000
Phone_Language	-0.20332921	0.23236957	0.39953830	0.091369362
Phone_Keyboard	-0.09209405	-0.10670909	0.15476629	0.250074284
Interaction_Score	-0.15586110	0.12059989	0.01950151	-0.004667569
Speaking_Score	-0.10542734	-0.03258461	0.16237217	0.170806905
Speak_Hours_Score	-0.11565422	-0.09014495	0.13302909	0.115316461
Use_French_Score	-0.11254932	-0.20216157	0.06748862	0.267523188
French_Hours_Score	-0.11949822	-0.07471738	0.03288211	0.177458375
Competence_Score	-0.02708528	0.03740722	0.64046015	0.203140460
Intention_Score	0.02918955	0.05172138	0.55474606	0.320380189
	Phone_Language	Phone_Keyboard	Interaction_Score	
Year_2024	-0.20332921	-0.092094053	-0.155861101	
Year_2025	0.23236957	-0.106709095	0.120599891	
Class_Level	0.39953830	0.154766293	0.019501506	
Host_Family	0.09136936	0.250074284	-0.004667569	
Phone_Language	1.00000000	0.113570900	0.121905388	
Phone_Keyboard	0.11357090	1.000000000	0.004978598	
Interaction_Score	0.12190539	0.004978598	1.000000000	
Speaking_Score	0.13127652	0.062204902	0.341816143	
Speak_Hours_Score	0.18368442	0.142494918	0.382297419	
Use_French_Score	0.12765385	0.264045353	0.359905285	
French_Hours_Score	0.14810030	0.145139401	0.358188934	
Competence_Score	0.40100917	0.030358879	0.261325495	

Intention_Score	0.46326215	0.158497209	0.345346548
	Speaking_Score	Speak_Hours_Score	Use_French_Score
Year_2024	-0.10542734	-0.11565422	-0.11254932
Year_2025	-0.03258461	-0.09014495	-0.20216157
Class_Level	0.16237217	0.13302909	0.06748862
Host_Family	0.17080691	0.11531646	0.26752319
Phone_Language	0.13127652	0.18368442	0.12765385
Phone_Keyboard	0.06220490	0.14249492	0.26404535
Interaction_Score	0.34181614	0.38229742	0.35990529
Speaking_Score	1.00000000	0.54635401	0.62373616
Speak_Hours_Score	0.54635401	1.00000000	0.62260415
Use_French_Score	0.62373616	0.62260415	1.00000000
French_Hours_Score	0.40418963	0.81685749	0.60418463
Competence_Score	0.39315018	0.33259247	0.31011951
Intention_Score	0.30832030	0.34507566	0.16272657
	French_Hours_Score	Competence_Score	Intention_Score
Year_2024	-0.11949822	-0.02708528	0.02918955
Year_2025	-0.07471738	0.03740722	0.05172138
Class_Level	0.03288211	0.64046015	0.55474606
Host_Family	0.17745838	0.20314046	0.32038019
Phone_Language	0.14810030	0.40100917	0.46326215
Phone_Keyboard	0.14513940	0.03035888	0.15849721
Interaction_Score	0.35818893	0.26132550	0.34534655
Speaking_Score	0.40418963	0.39315018	0.30832030
Speak_Hours_Score	0.81685749	0.33259247	0.34507566
Use_French_Score	0.60418463	0.31011951	0.16272657
French_Hours_Score	1.00000000	0.19251489	0.27018609
Competence_Score	0.19251489	1.00000000	0.56004456
Intention_Score	0.27018609	0.56004456	1.00000000

Correlation strength will be defined as:

- 0 to abs(0.2) - Very weak or no association
- abs(0.2) to abs(0.4) - Weak association
- abs(0.4) to abs(0.6) - Moderation association
- abs(0.6) to abs(0.8) - Strong association
- abs(0.8) to abs(1.0) - Perfect or very strong association

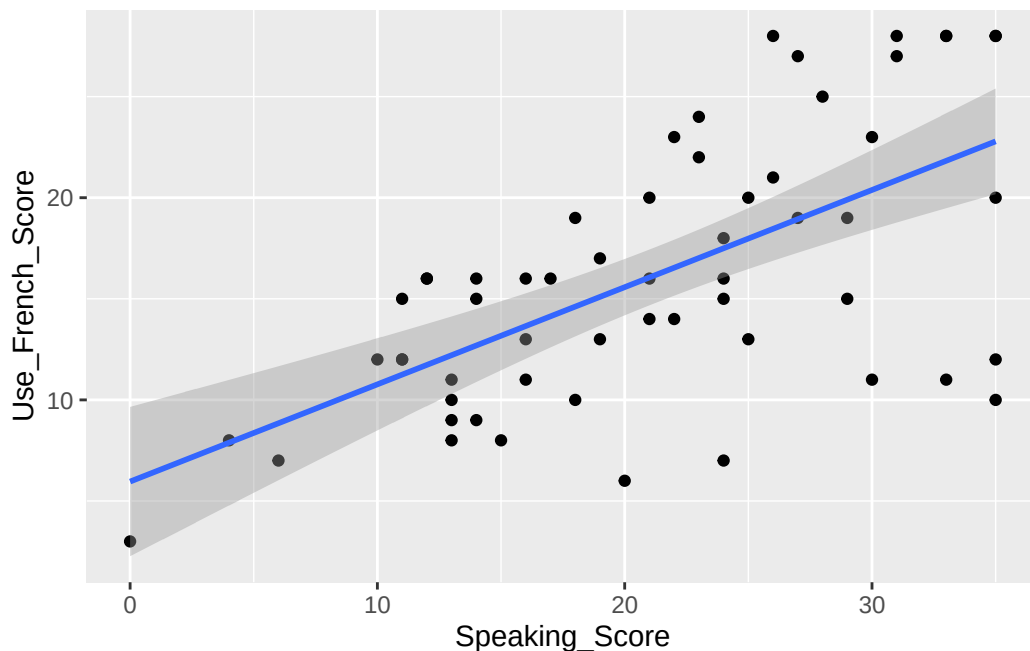
The variables with nearly zero or very weak linear correlation with Competence_Score were: Year_2024, Year_2025, Phone_Keyboard, and French_Hours_Score. The ones with weak linear correlation were: Host_Family, Interaction_Score, Speaking_Score,

Speak_Hours_Score, and Use_French_Score. The variables with moderate linear correlation were: Phone_Language and Intention_Score. Finally the only variable with a strong association was Class_Level.

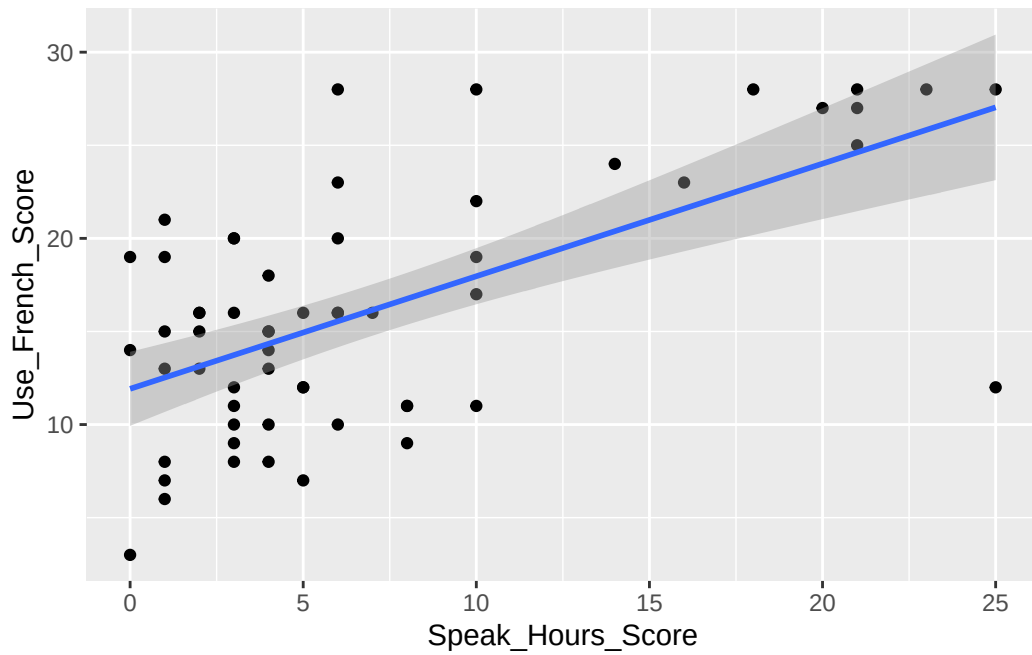
The variables with nearly zero or very weak linear correlation with Intention_Score were: Year_2024, Year_2025, Phone_Keyboard, and Use_French_Score. The variables with weak linear correlation were: Host_Family, Interaction_Score, Speaking_Score, Speak_Hours_Score, and French_Hours_Score. The variables with moderate linear correlation were: Class_Level, Phone_Language, and Competence_Score.

Speaking_Score has a strong association with Use_French_Score. Speak_Hours_Score has a strong association with Use_French_Score and a very strong association with French_Hours_Score. Finally, Use_French_Score also has a strong association with French_Hours_Score. It's possible that this may cause some of these variables to have too high of multicollinearity that will affect the model fit and whether they are statistically significant.

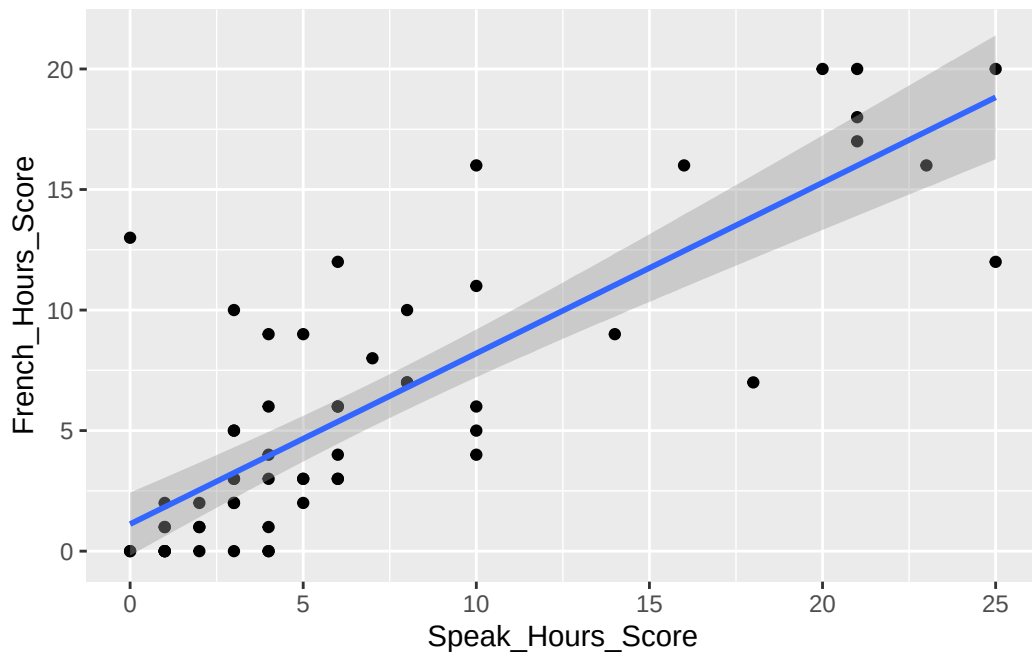
```
ggplot(french_data, aes(Speaking_Score, Use_French_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



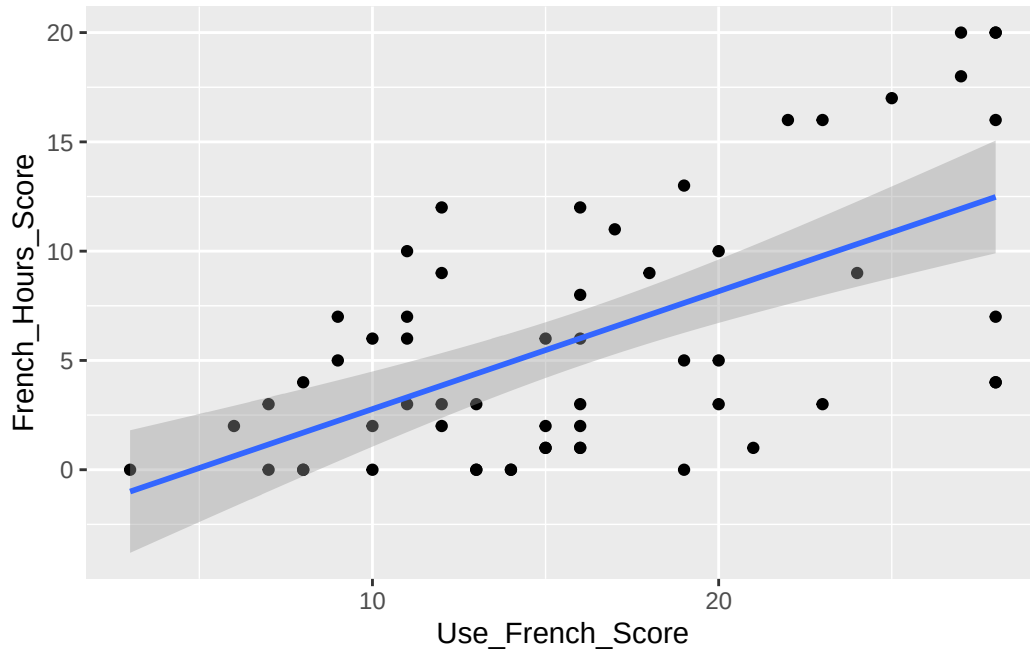
```
ggplot(french_data, aes(Speak_Hours_Score, Use_French_Score)) +  
  geom_point() +  
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



```
ggplot(french_data, aes(Speak_Hours_Score, French_Hours_Score)) +
  geom_point() +
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



```
ggplot(french_data, aes(Use_French_Score, French_Hours_Score)) +
  geom_point() +
  geom_smooth(method = 'lm', formula = 'y ~ x')
```



Linear Models

The Screen_Time variable had several issues with data quality, and along with the various details found above, our client determined that it had too many issues. Therefore, the variable was not included in the model.

```
competence_model <- lm(Competence_Score ~ Year_2024 + Year_2025 + Class_Level +
  Host_Family + Phone_Language +
  Phone_Keyboard + Interaction_Score +
  Speaking_Score + Speak_Hours_Score +
  Use_French_Score + French_Hours_Score,
  data = french_data)

summary(competence_model)
```

Call:

```
lm(formula = Competence_Score ~ Year_2024 + Year_2025 + Class_Level +  
    Host_Family + Phone_Language + Phone_Keyboard + Interaction_Score +  
    Speaking_Score + Speak_Hours_Score + Use_French_Score + French_Hours_Score,  
    data = french_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-30.5088	-4.1554	-0.1637	7.6940	30.7992

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.5695	10.3537	1.311	0.197
Year_20241	-0.2997	4.6182	-0.065	0.949
Year_20251	0.8160	4.2773	0.191	0.850
Class_Level	8.1887	1.6903	4.844	1.48e-05 ***
Host_Family1	-1.2939	4.0917	-0.316	0.753
Phone_Language1	4.2433	3.9690	1.069	0.291
Phone_Keyboard1	-5.5604	5.0987	-1.091	0.281
Interaction_Score	0.4249	0.3891	1.092	0.281
Speaking_Score	0.2545	0.2767	0.920	0.362
Speak_Hours_Score	0.3787	0.4920	0.770	0.445
Use_French_Score	0.4457	0.4144	1.075	0.288
French_Hours_Score	-0.4052	0.5356	-0.757	0.453

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.91 on 46 degrees of freedom

Multiple R-squared: 0.5629, Adjusted R-squared: 0.4584

F-statistic: 5.385 on 11 and 46 DF, p-value: 2.011e-05

```
# Check the variance inflation factor  
vif(competence_model)
```

Year_2024	Year_2025	Class_Level	Host_Family
1.359156	1.402506	1.480306	1.394276
Phone_Language	Phone_Keyboard	Interaction_Score	Speaking_Score
1.364036	1.186039	1.289915	1.949006
Speak_Hours_Score	Use_French_Score	French_Hours_Score	
3.880456	2.597527	3.455914	

A VIF of 1 - 5 indicates low to moderate multicollinearity, and since none of the VIFs were

above 5, the model fit is likely not being affected by it.

```
intention_model <- lm(Intention_Score ~ Year_2024 + Year_2025 +  
                      Class_Level + Host_Family +  
                      Phone_Language + Phone_Keyboard +  
                      Interaction_Score + Speaking_Score + Speak_Hours_Score +  
                      Use_French_Score + French_Hours_Score,  
                      data = french_data)  
  
summary(intention_model)
```

Call:

```
lm(formula = Intention_Score ~ Year_2024 + Year_2025 + Class_Level +  
    Host_Family + Phone_Language + Phone_Keyboard + Interaction_Score +  
    Speaking_Score + Speak_Hours_Score + Use_French_Score + French_Hours_Score,  
    data = french_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.5129	-2.5545	0.8018	2.5540	6.4883

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.30035	3.05011	-1.082	0.28487
Year_20241	0.86234	1.36047	0.634	0.52932
Year_20251	-0.06608	1.26005	-0.052	0.95841
Class_Level	1.38493	0.49795	2.781	0.00782 **
Host_Family1	1.69209	1.20537	1.404	0.16710
Phone_Language1	2.67133	1.16922	2.285	0.02699 *
Phone_Keyboard1	1.05784	1.50202	0.704	0.48481
Interaction_Score	0.28903	0.11464	2.521	0.01522 *
Speaking_Score	0.09929	0.08151	1.218	0.22942
Speak_Hours_Score	0.11869	0.14493	0.819	0.41703
Use_French_Score	-0.22831	0.12209	-1.870	0.06786 .
French_Hours_Score	0.05871	0.15780	0.372	0.71155

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.803 on 46 degrees of freedom

Multiple R-squared: 0.5494, Adjusted R-squared: 0.4416

F-statistic: 5.098 on 11 and 46 DF, p-value: 3.661e-05

```
# Check the variance inflation factor
vif(intention_model)
```

Year_2024	Year_2025	Class_Level	Host_Family
1.359156	1.402506	1.480306	1.394276
Phone_Language	Phone_Keyboard	Interaction_Score	Speaking_Score
1.364036	1.186039	1.289915	1.949006
Speak_Hours_Score	Use_French_Score	French_Hours_Score	
3.880456	2.597527	3.455914	

Similar to above, none of the VIFs were above 5, therefore the model fit is likely not being affected by multicollinearity.

Feature Selection using BIC

BIC is the Bayesian Information Criterion which can be used to select which model is fitting the data better based on which has the lowest BIC. It has a stronger penalty for models with more parameters. The equation is $BIC = -2 \cdot \ln(\text{Likelihood}) + k \cdot \ln(n)$, where k is the number of parameters in the model and n is the number of observations in the dataset. We decided to use BIC because we wanted to obtain a more parsimonious model. The feature selection method uses a backward elimination approach which removes one variable at a time and chooses the model that reduced the BIC the most. Backward elimination is the default method for performing feature selection in this way. Using forward or both as the method for selection provided the same results for both models.

Competence Model

```
# Feature selection using backwards elimination and BIC
# k is what defines the model the be BIC even though the output says AIC
competence_model_reduced <- step(competence_model, method = "backward",
                                k = log(nrow(competence_model$model)))
```

Start: AIC=332.01

```
Competence_Score ~ Year_2024 + Year_2025 + Class_Level + Host_Family +
  Phone_Language + Phone_Keyboard + Interaction_Score + Speaking_Score +
  Speak_Hours_Score + Use_French_Score + French_Hours_Score
```

Df	Sum of Sq	RSS	AIC
----	-----------	-----	-----

- Year_2024	1	0.7	7666.9	327.95
- Year_2025	1	6.1	7672.3	327.99
- Host_Family	1	16.7	7682.9	328.07
- French_Hours_Score	1	95.4	7761.6	328.66
- Speak_Hours_Score	1	98.8	7765.0	328.69
- Speaking_Score	1	141.0	7807.3	329.00
- Phone_Language	1	190.5	7856.7	329.37
- Use_French_Score	1	192.7	7858.9	329.38
- Phone_Keyboard	1	198.2	7864.4	329.43
- Interaction_Score	1	198.7	7864.9	329.43
<none>			7666.2	332.01
- Class_Level	1	3911.2	11577.4	351.85

Step: AIC=327.95

Competence_Score ~ Year_2025 + Class_Level + Host_Family + Phone_Language +
 Phone_Keyboard + Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score + French_Hours_Score

	Df	Sum of Sq	RSS	AIC
- Year_2025	1	8.8	7675.7	323.96
- Host_Family	1	18.5	7685.4	324.03
- French_Hours_Score	1	95.0	7761.9	324.60
- Speak_Hours_Score	1	98.7	7765.6	324.63
- Speaking_Score	1	141.8	7808.7	324.95
- Use_French_Score	1	197.1	7864.0	325.36
- Phone_Language	1	197.7	7864.6	325.37
- Phone_Keyboard	1	199.0	7866.0	325.38
- Interaction_Score	1	199.3	7866.2	325.38
<none>			7666.9	327.95
- Class_Level	1	3961.3	11628.2	348.05

Step: AIC=323.96

Competence_Score ~ Class_Level + Host_Family + Phone_Language +
 Phone_Keyboard + Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score + French_Hours_Score

	Df	Sum of Sq	RSS	AIC
- Host_Family	1	22.1	7697.8	320.06
- French_Hours_Score	1	91.9	7767.6	320.59
- Speak_Hours_Score	1	94.6	7770.3	320.61
- Speaking_Score	1	151.6	7827.3	321.03
- Use_French_Score	1	188.3	7864.0	321.30
- Phone_Keyboard	1	200.6	7876.3	321.39

- Interaction_Score	1	219.5	7895.3	321.53
- Phone_Language	1	239.1	7914.8	321.67
<none>			7675.7	323.96
- Class_Level	1	3953.6	11629.3	343.99

Step: AIC=320.06

Competence_Score ~ Class_Level + Phone_Language + Phone_Keyboard +
 Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score + French_Hours_Score

	Df	Sum of Sq	RSS	AIC
- French_Hours_Score	1	110.9	7808.7	316.83
- Speak_Hours_Score	1	115.4	7813.1	316.86
- Speaking_Score	1	147.4	7845.2	317.10
- Use_French_Score	1	171.3	7869.1	317.28
- Phone_Keyboard	1	225.7	7923.5	317.68
- Interaction_Score	1	234.0	7931.8	317.74
- Phone_Language	1	254.3	7952.1	317.89
<none>			7697.8	320.06
- Class_Level	1	4306.8	12004.6	341.77

Step: AIC=316.83

Competence_Score ~ Class_Level + Phone_Language + Phone_Keyboard +
 Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Speak_Hours_Score	1	21.3	7830.0	312.93
- Use_French_Score	1	115.9	7924.5	313.62
- Speaking_Score	1	209.0	8017.7	314.30
- Interaction_Score	1	213.3	8022.0	314.33
- Phone_Keyboard	1	224.6	8033.3	314.42
- Phone_Language	1	241.5	8050.2	314.54
<none>			7808.7	316.83
- Class_Level	1	4516.5	12325.1	339.24

Step: AIC=312.93

Competence_Score ~ Class_Level + Phone_Language + Phone_Keyboard +
 Interaction_Score + Speaking_Score + Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Use_French_Score	1	184.4	8014.3	310.22
- Phone_Keyboard	1	224.1	8054.0	310.50

- Interaction_Score	1	246.1	8076.1	310.66
- Speaking_Score	1	250.8	8080.8	310.70
- Phone_Language	1	255.2	8085.2	310.73
<none>			7830.0	312.93
- Class_Level	1	4556.1	12386.0	335.47

Step: AIC=310.22

Competence_Score ~ Class_Level + Phone_Language + Phone_Keyboard +
Interaction_Score + Speaking_Score

	Df	Sum of Sq	RSS	AIC
- Phone_Keyboard	1	129.3	8143.7	307.09
- Phone_Language	1	277.3	8291.6	308.13
- Interaction_Score	1	356.7	8371.0	308.68
<none>			8014.3	310.22
- Speaking_Score	1	849.2	8863.5	312.00
- Class_Level	1	4421.5	12435.8	331.64

Step: AIC=307.09

Competence_Score ~ Class_Level + Phone_Language + Interaction_Score +
Speaking_Score

	Df	Sum of Sq	RSS	AIC
- Phone_Language	1	257.2	8400.9	304.83
- Interaction_Score	1	364.4	8508.1	305.57
<none>			8143.7	307.09
- Speaking_Score	1	825.4	8969.1	308.63
- Class_Level	1	4306.9	12450.6	327.65

Step: AIC=304.83

Competence_Score ~ Class_Level + Interaction_Score + Speaking_Score

	Df	Sum of Sq	RSS	AIC
- Interaction_Score	1	437.2	8838.1	303.71
<none>			8400.9	304.83
- Speaking_Score	1	856.9	9257.8	306.40
- Class_Level	1	6106.4	14507.3	332.46

Step: AIC=303.71

Competence_Score ~ Class_Level + Speaking_Score

	Df	Sum of Sq	RSS	AIC
<none>			8838.1	303.71

```
- Speaking_Score 1      1506.1 10344.2 308.78
- Class_Level    1      5989.2 14827.3 329.66
```

```
summary(competence_model_reduced)
```

Call:

```
lm(formula = Competence_Score ~ Class_Level + Speaking_Score,
    data = french_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-32.922	-3.892	0.666	7.679	27.324

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	20.1972	4.6911	4.305	6.92e-05	***
Class_Level	8.4405	1.3826	6.105	1.09e-07	***
Speaking_Score	0.6038	0.1972	3.061	0.0034	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.68 on 55 degrees of freedom

Multiple R-squared: 0.4961, Adjusted R-squared: 0.4777

F-statistic: 27.07 on 2 and 55 DF, p-value: 6.536e-09

Using backward elimination based on BIC, we get a final reduced model with Class_Level and Speaking_Score. However, this model rejected normality of the residuals in the Shapiro-Wilks test, which means the model assumptions were not met and the model was not valid. The BIC difference between the final model and the one above it was only 1.12, which is very small. Therefore we chose to use the model with Interaction_Score.

```
# Reverting to the second best BIC model to meet model assumptions
competence_model_reduced <- lm(Competence_Score ~
    Class_Level +
    Interaction_Score +
    Speaking_Score,
    data = french_data)

summary(competence_model_reduced)
```

```
Call:
lm(formula = Competence_Score ~ Class_Level + Interaction_Score +
    Speaking_Score, data = french_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-34.719	-5.057	0.283	8.142	31.192

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.5654	8.3338	1.028	0.3086
Class_Level	8.5291	1.3614	6.265	6.39e-08 ***
Interaction_Score	0.5910	0.3525	1.676	0.0995 .
Speaking_Score	0.4849	0.2066	2.347	0.0226 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

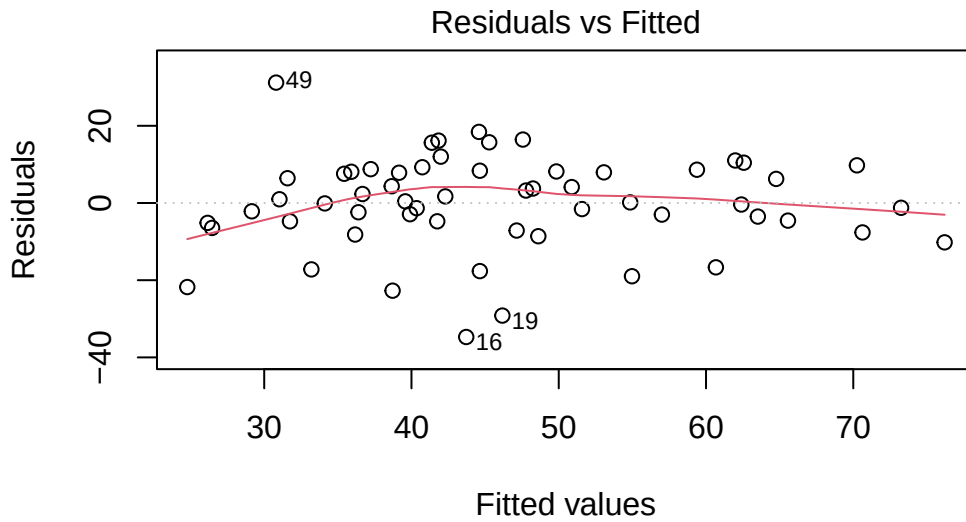
Residual standard error: 12.47 on 54 degrees of freedom

Multiple R-squared: 0.521, Adjusted R-squared: 0.4944

F-statistic: 19.58 on 3 and 54 DF, p-value: 1.02e-08

In this model the Class_Level and Speaking_Score variables are statistically significant at a significance level of 0.05. Even though the Interaction_Score is not statistically significant at a significance level of 0.05, including it in the model allows the model to meet model assumptions.

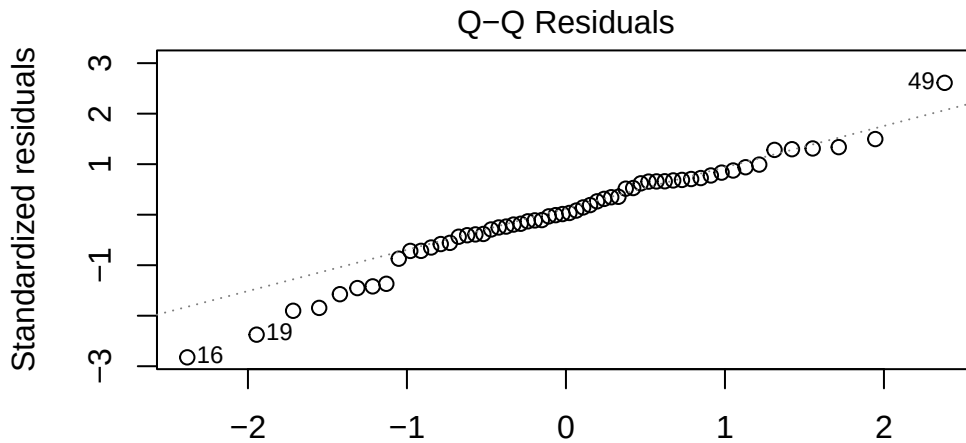
```
# Residuals vs Fitted
plot(competence_model_reduced, which = 1)
```



`lm(Competence_Score ~ Class_Level + Interaction_Score + Speaking_Sc`

Generally the residuals are spread out evenly across the fitted values, showing that there is roughly equal variance of the residuals. However, the three outliers identified by the plot are observations 16, 19, and 49.

```
# QQ Plot
plot(competence_model_reduced, which = 2)
```



Theoretical Quantiles

`lm(Competence_Score ~ Class_Level + Interaction_Score + Speaking_Sc`

The standardized residuals only slightly deviate from the theoretical normal quantiles at the ends, indicating that the residuals are likely normally distributed.

```
shapiro.test(resid(competence_model_reduced))
```

Shapiro-Wilk normality test

```
data:  resid(competence_model_reduced)
W = 0.9685, p-value = 0.1359
```

The Shapiro-Wilk test tests the null hypothesis that a sample comes from a normally distributed population. With a p-value of 0.1359, this model fails to reject the null hypothesis at a significance level of 0.05. Therefore, there is evidence to support that the sample does come from a normally distributed population.

```
# Once again we can check for multicollinearity
vif(competence_model_reduced)
```

Class_Level	Interaction_Score	Speaking_Score
1.028629	1.134005	1.164269

All of the VIFs are very low which indicates that there is very little multicollinearity.

Looking at the outliers from the plots above, observations 16 and 19 have large negative residuals and 49 has a large positive one. A residual is the actual value minus the predicted one. So a negative residual means the model overestimated, and a positive means it underestimated.

```
french_data |>
  mutate(Observation = seq(1:nrow(french_data))) |>
  dplyr::select(Class_Level, Interaction_Score, Speaking_Score,
                Competence_Score, Observation) |>
  arrange(Competence_Score)
```

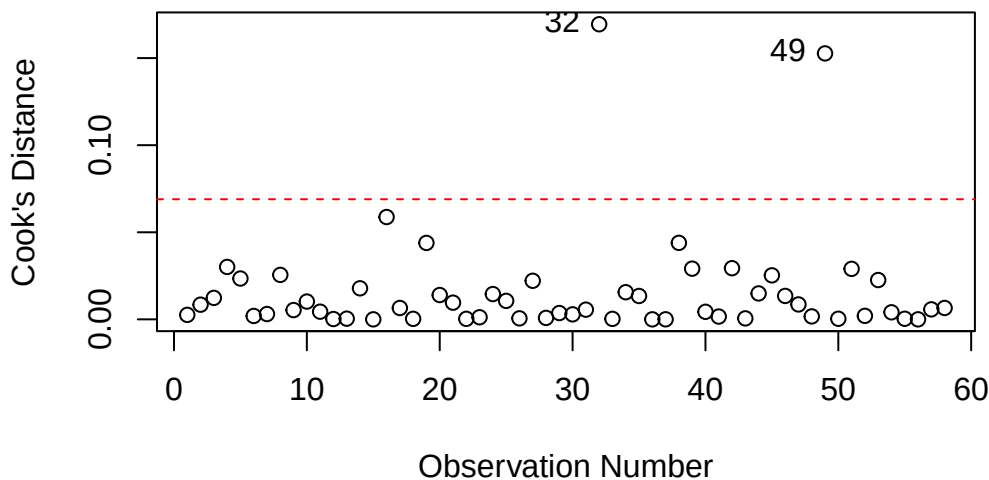
```
# A tibble: 58 x 5
```

	Class_Level	Interaction_Score	Speaking_Score	Competence_Score	Observation
	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	1	13	0	3	32
2	1	27	22	9	16
3	0	22	24	16	42
4	1	21	19	16	53
5	2	20	18	17	19
6	0	27	4	20	3
7	0	15	18	21	9
8	0	22	21	27	6
9	1	22	30	27	39
10	0	25	12	27	43

```
# i 48 more rows
```

As can be seen from the data, observations 16 and 19 had relatively low Competence_Scores given their Class_Level, Interaction_Score, and Speaking_Score. On the other hand observation 49 has a surprisingly high Competence_Score given it's relatively low Class_Level, Interaction_Score, and Speaking_Score.

```
cooksdc <- cooks.distance(competence_model_reduced)
plot(cooksdc, xlab = "Observation Number", ylab = "Cook's Distance")
abline(h = 4/nrow(french_data), col = "red", lty = 2)
inflabc <- which(cooksdc > 4/nrow(french_data))
text(inflabc, cooksdc[inflabc], labels = inflabc, pos=2)
```



Looking at Cook's Distance we see that 32 and 49 are both influential values. Observation 49 was discussed above, but as can be seen observation 32 is the value with the lowest Competence_Score as well as the Speaking_Score being 0.

Intention Model

```
# Feature selection using backwards elimination and BIC
# k is what defines the model the be BIC even though the output says AIC
intention_model_reduced <- step(intention_model, method = "backwards",
                                k = log(nrow(competence_model$model)))
```

Start: AIC=190.23

```
Intention_Score ~ Year_2024 + Year_2025 + Class_Level + Host_Family +
  Phone_Language + Phone_Keyboard + Interaction_Score + Speaking_Score +
  Speak_Hours_Score + Use_French_Score + French_Hours_Score
```

	Df	Sum of Sq	RSS	AIC
- Year_2025	1	0.040	665.34	186.18
- French_Hours_Score	1	2.002	667.31	186.35
- Year_2024	1	5.811	671.12	186.68
- Phone_Keyboard	1	7.174	672.48	186.79

- Speak_Hours_Score	1	9.700	675.00	187.01
- Speaking_Score	1	21.458	686.76	188.01
- Host_Family	1	28.501	693.81	188.61
<none>			665.30	190.23
- Use_French_Score	1	50.574	715.88	190.42
- Phone_Language	1	75.497	740.80	192.41
- Interaction_Score	1	91.935	757.24	193.68
- Class_Level	1	111.876	777.18	195.19

Step: AIC=186.18

Intention_Score ~ Year_2024 + Class_Level + Host_Family + Phone_Language +
Phone_Keyboard + Interaction_Score + Speaking_Score + Speak_Hours_Score +
Use_French_Score + French_Hours_Score

	Df	Sum of Sq	RSS	AIC
- French_Hours_Score	1	1.980	667.32	182.29
- Year_2024	1	7.098	672.44	182.73
- Phone_Keyboard	1	7.282	672.63	182.75
- Speak_Hours_Score	1	9.846	675.19	182.97
- Speaking_Score	1	21.472	686.82	183.96
- Host_Family	1	28.659	694.00	184.56
<none>			665.34	186.18
- Use_French_Score	1	52.901	718.25	186.55
- Phone_Language	1	77.953	743.30	188.54
- Interaction_Score	1	93.404	758.75	189.74
- Class_Level	1	111.953	777.30	191.14

Step: AIC=182.29

Intention_Score ~ Year_2024 + Class_Level + Host_Family + Phone_Language +
Phone_Keyboard + Interaction_Score + Speaking_Score + Speak_Hours_Score +
Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Year_2024	1	6.768	674.09	178.81
- Phone_Keyboard	1	7.035	674.36	178.84
- Speaking_Score	1	19.732	687.06	179.92
- Host_Family	1	32.332	699.66	180.97
- Speak_Hours_Score	1	35.970	703.29	181.27
<none>			667.32	182.29
- Use_French_Score	1	50.947	718.27	182.50
- Phone_Language	1	79.126	746.45	184.73
- Interaction_Score	1	96.147	763.47	186.03
- Class_Level	1	110.025	777.35	187.08

Step: AIC=178.81

Intention_Score ~ Class_Level + Host_Family + Phone_Language +
 Phone_Keyboard + Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Phone_Keyboard	1	5.454	679.55	175.22
- Speaking_Score	1	18.435	692.53	176.32
- Speak_Hours_Score	1	35.949	710.04	177.77
- Host_Family	1	40.425	714.52	178.13
<none>			674.09	178.81
- Use_French_Score	1	51.983	726.08	179.06
- Phone_Language	1	72.615	746.71	180.69
- Interaction_Score	1	92.540	766.63	182.21
- Class_Level	1	121.093	795.19	184.34

Step: AIC=175.22

Intention_Score ~ Class_Level + Host_Family + Phone_Language +
 Interaction_Score + Speaking_Score + Speak_Hours_Score +
 Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Speaking_Score	1	15.990	695.54	172.51
- Speak_Hours_Score	1	36.414	715.96	174.19
- Host_Family	1	45.831	725.38	174.95
- Use_French_Score	1	46.946	726.49	175.03
<none>			679.55	175.22
- Phone_Language	1	74.784	754.33	177.22
- Interaction_Score	1	89.977	769.52	178.37
- Class_Level	1	125.844	805.39	181.01

Step: AIC=172.51

Intention_Score ~ Class_Level + Host_Family + Phone_Language +
 Interaction_Score + Speak_Hours_Score + Use_French_Score

	Df	Sum of Sq	RSS	AIC
- Use_French_Score	1	32.669	728.21	171.11
- Host_Family	1	44.967	740.50	172.08
- Speak_Hours_Score	1	49.955	745.49	172.47
<none>			695.54	172.51
- Phone_Language	1	72.926	768.46	174.23
- Interaction_Score	1	100.350	795.89	176.27

```
- Class_Level      1    140.208 835.75 179.10
```

Step: AIC=171.11

```
Intention_Score ~ Class_Level + Host_Family + Phone_Language +  
  Interaction_Score + Speak_Hours_Score
```

	Df	Sum of Sq	RSS	AIC
- Speak_Hours_Score	1	21.878	750.08	168.77
- Host_Family	1	27.865	756.07	169.23
<none>			728.21	171.11
- Phone_Language	1	69.594	797.80	172.34
- Interaction_Score	1	83.146	811.35	173.32
- Class_Level	1	159.412	887.62	178.53

Step: AIC=168.77

```
Intention_Score ~ Class_Level + Host_Family + Phone_Language +  
  Interaction_Score
```

	Df	Sum of Sq	RSS	AIC
- Host_Family	1	32.835	782.92	167.19
<none>			750.08	168.77
- Phone_Language	1	79.425	829.51	170.54
- Interaction_Score	1	137.252	887.34	174.45
- Class_Level	1	165.405	915.49	176.26

Step: AIC=167.19

```
Intention_Score ~ Class_Level + Phone_Language + Interaction_Score
```

	Df	Sum of Sq	RSS	AIC
<none>			782.92	167.19
- Phone_Language	1	73.851	856.77	168.36
- Interaction_Score	1	136.572	919.49	172.46
- Class_Level	1	251.577	1034.50	179.29

```
summary(intention_model_reduced)
```

Call:

```
lm(formula = Intention_Score ~ Class_Level + Phone_Language +  
  Interaction_Score, data = french_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.0739	-2.7158	0.5275	2.5269	7.6146

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.1559	2.5199	-1.252	0.215823
Class_Level	1.8630	0.4472	4.166	0.000113 ***
Phone_Language1	2.4871	1.1020	2.257	0.028081 *
Interaction_Score	0.3127	0.1019	3.069	0.003355 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

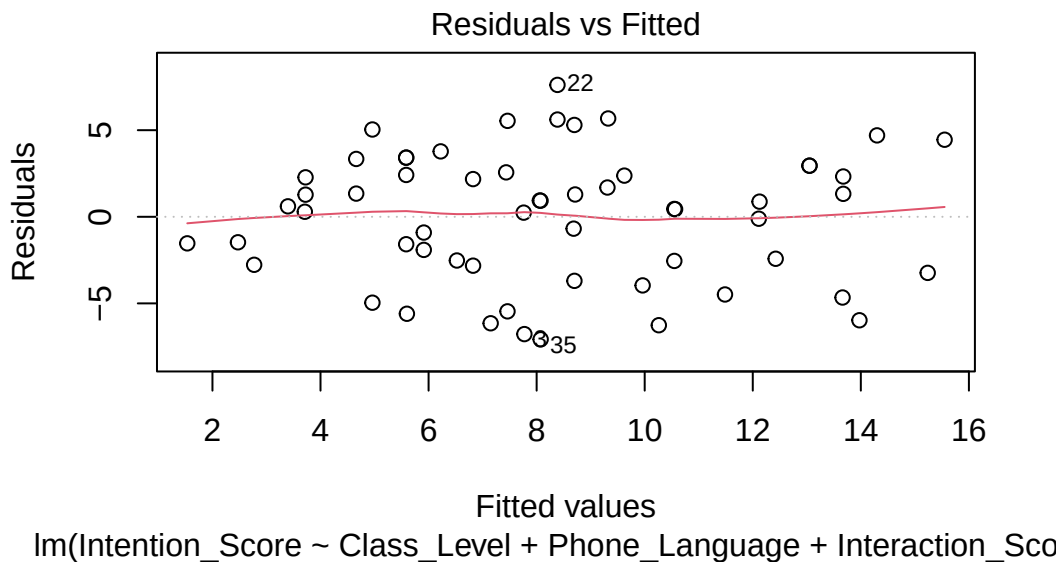
Residual standard error: 3.808 on 54 degrees of freedom

Multiple R-squared: 0.4697, Adjusted R-squared: 0.4403

F-statistic: 15.94 on 3 and 54 DF, p-value: 1.514e-07

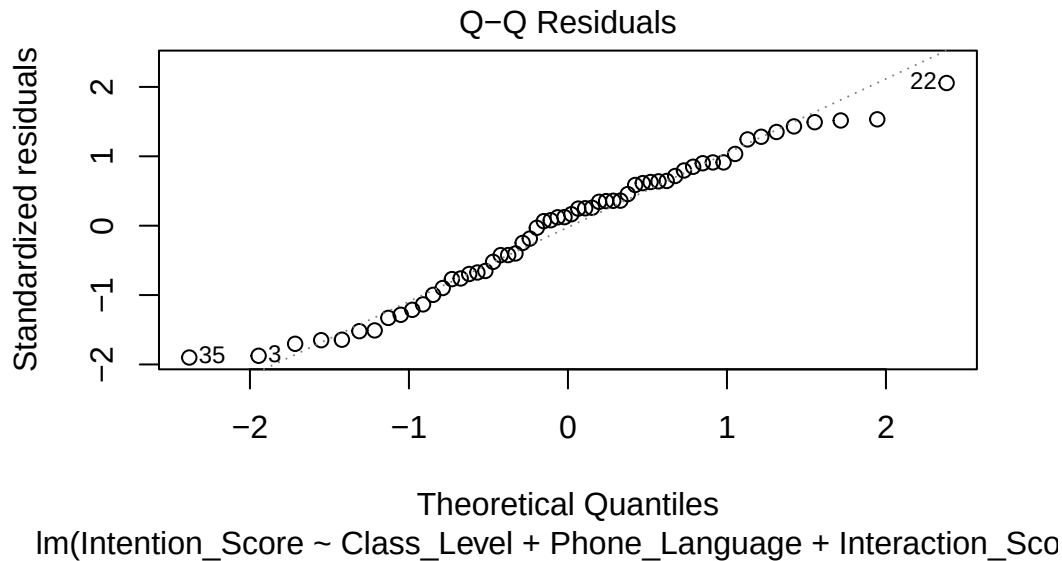
The final reduced model included Class_Level, Phone_Language, and Interaction_Score. All of these variables are statistically significant at a significance level of 0.05.

```
# Residuals vs Fitted
plot(intention_model_reduced, which = 1)
```



The residuals appear to be roughly evenly spaced across the fitted values, indicating there is equal variance of the residuals. The three outliers identified by the plot are observations 3, 22, and 35.

```
# QQ Plot  
plot(intention_model_reduced, which = 2)
```



The standardized residuals appear to align roughly with the theoretical normal quantiles, indicating that the residuals are normally distributed.

```
shapiro.test(resid(intention_model_reduced))
```

Shapiro-Wilk normality test

```
data: resid(intention_model_reduced)  
W = 0.97178, p-value = 0.1943
```

With a p-value of 0.1943, this model fails to reject the null hypothesis at a significance level of 0.05. Therefore, there is evidence to support that the sample does come from a normally distributed population.

```
# Once again we can check for multicollinearity
vif(intention_model_reduced)
```

```
Class_Level    Phone_Language Interaction_Score
1.191180        1.208690        1.016132
```

All of the VIFs are very low which indicates that there is very little multicollinearity.

Looking at the outliers from the plots above, observations 3 and 35 have large negative residuals and 22 has a large positive one. A residual is the actual value - the predicted one. So a negative residual means the model overestimated, and a positive means it underestimated.

```
french_data |>
  mutate(Observation = seq(1:nrow(french_data))) |>
  dplyr::select(Class_Level, Phone_Language, Interaction_Score,
                Intention_Score, Observation) |>
  arrange(Intention_Score)
```

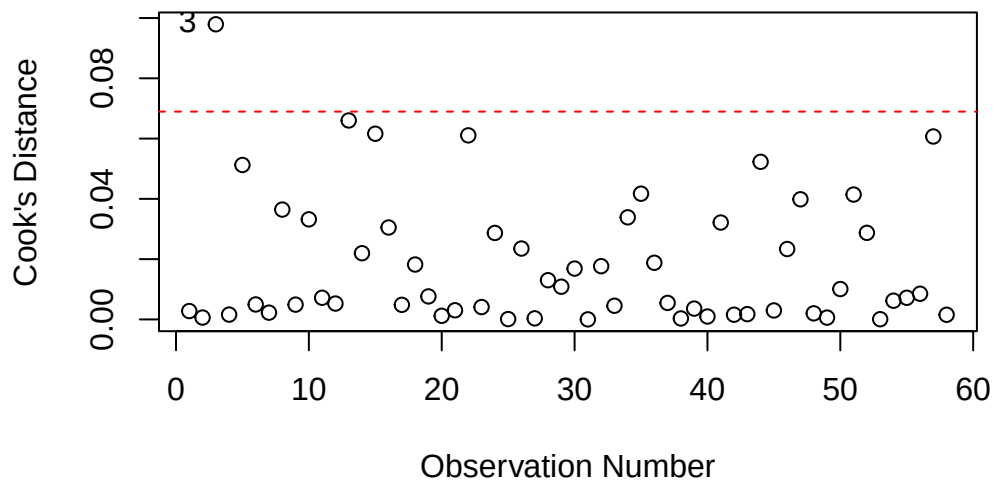
```
# A tibble: 58 x 5
  Class_Level Phone_Language Interaction_Score Intention_Score Observation
    <dbl>    <fct>          <dbl>          <dbl>          <int>
1         0 0              15              0              9
2         1 0              20              0             18
3         1 0              13              0             32
4         0 0              28              0             47
5         0 1              27              1              3
6         1 0              27              1             16
7         2 0              24              1             35
8         0 0              18              1             45
9         0 1              26              2             15
10        0 0              29              4             12
# i 48 more rows
```

While it's somewhat hard to see it appears that the Intention_Score for observation 3 and 35 were lower than the model expected given their Class_Level, Phone_Language, and Interaction_Score combination. With observation 22 it is easier to see that it has a higher Intention_Score than the model expected given a Class_Level of 1, as well as the Phone_Language and Interaction_Score variables.

```

cooksdi <- cooks.distance(intention_model_reduced)
plot(cooksdi, xlab = "Observation Number", ylab = "Cook's Distance")
abline(h = 4/nrow(french_data), col = "red", lty = 2)
inflabi <- which(cooksdi > 4/nrow(french_data))
text(inflabi, cooksdi[inflabi], labels = inflabi, pos=2)

```



Observation 3 is an influential point and it was described above. However as discussed before, since they are valid data points, they were not removed from the model.